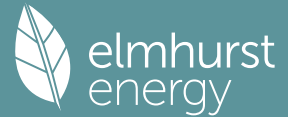


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Plot Reference	CPG003338 - Planning				Issued on Date	15/06/2026
Assessment Reference	Plot 1	Plot Type Ref	Barn E Plot 1			
Plot Address	Barn E at Gwamnik Mill Farm, St Allen, Truro, Cornwall, TR4 9QU				SAP Version	10.2
SAP Rating	95 A	DER	0.25	TER	10.10	
Environmental	100 A	% DER < TER				97.52
CO ₂ Emissions (t/year)	-0.07	DFEE	47.99	TFEE	55.81	
Compliance Check	See BREL	% DFEE < TFEE				14.02
% DPER < TPER	76.90	DPER	12.79	TPER	55.39	
Assessor Details	Mr. Aaron Roberts				Assessor ID	DV31-0001
Client	Mr & Mrs Trevor Burley, Mr & Mrs Trevor Burley					

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF DWELLING EMISSIONS FOR REGULATIONS COMPLIANCE

1. Overall dwelling characteristics

	Area (m ²)	Storey height (m)	Volume (m ³)
Main dwelling			
Ground floor	134.2700 (1b)	3.3400 (2b)	448.4618 (1b) - (3b)
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700		448.4618 (4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n)	448.4618 (5)

2. Ventilation rate

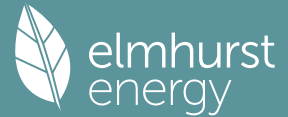
	m ³ per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	1 * 10 = 10.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	0 * 10 = 0.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	10.0000 / (5) = 0.0223 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	1.0000 (17)
Infiltration rate	0.0723 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.0723 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.0922	0.0904	0.0886	0.0795	0.0777	0.0687	0.0687	0.0669	0.0723	0.0777	0.0813	0.0850 (22b)
Balanced mechanical ventilation with heat recovery												
If mechanical ventilation												0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												84.6000 (23c)
Effective ac	0.1692	0.1674	0.1656	0.1565	0.1547	0.1457	0.1457	0.1439	0.1493	0.1547	0.1583	0.1620 (25)

3. Heat losses and heat loss parameter

Element	Gross m ²	Openings m ²	NetArea m ²	U-value W/m ² K	A x U W/K	K-value kJ/m ² K	A x K kJ/K
Main dwelling							
Door			4.4500	0.9615	4.2788		(27)
Window			20.9300	0.9615	20.1250		(27)
Door			1.7100	1.0000	1.7100		(26)
Ground Floor			134.2700	0.1100	14.7697	110.0000	14769.7000 (28a)
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1400	15.3328	9.0000	985.6800 (29a)
External Roof 1 - Pitched	155.0400		155.0400	0.1400	21.7056	9.0000	1395.3600 (30)
Total net area of external elements Aum(A, m ²)			425.9200				(31)
Fabric heat loss, W/K = Sum (A x U)					(26)...(30) + (32) = 77.9219		(33)
Main dwelling							
Party Wall 1			5.2800	0.0000	0.0000	20.0000	105.6000 (32)
Internal Wall 1 - GF			242.5100			9.0000	2182.5900 (32c)
Heat capacity Cm = Sum(A x k)							(28)...(30) + (32) + (32a)...(32e) = 19438.9300 (34)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m ² K							144.7749 (35)
List of Thermal Bridges							

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K1 Element	Length	Psi-value	Total
E2 Other lintels (including other steel lintels)	19.1200	0.0840	1.6061
E3 Sill	10.8000	0.0340	0.3672
E4 Jamb	36.2000	0.0430	1.5566
E5 Ground floor (normal)	54.4600	0.0210	1.1437
E11 Eaves (insulation at rafter level)	43.4400	0.0400	1.7376
E13 Gable (insulation at rafter level)	12.7200	0.0240	0.3053
E16 Corner (normal)	7.6500	0.0300	0.2295
R4 Ridge (vaulted ceiling)	24.3600	0.1200	2.9232
E18 Party wall between dwellings	5.1000	0.0400	0.2040
P1 Party wall - Ground floor	5.2800	0.3200	1.6896
P5 Party wall - Roof (insulation at rafter level)	6.3600	0.4800	3.0528
Thermal bridges (Sum(L x Psi) calculated using Appendix K)			14.8155 (36)
Point Thermal bridges		(36a) =	0.0000
Total fabric heat loss		(33) + (36) + (36a) =	92.7375 (37)

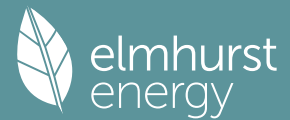
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)												
(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Heat transfer coeff	25.0374	24.7699	24.5024	23.1650	22.8975	21.5601	21.5601	21.2926	22.0950	22.8975	23.4325	23.9675 (38)
Average = Sum(39)m / 12 =	117.7749	117.5074	117.2399	115.9025	115.6350	114.2975	114.2975	114.0300	114.8325	115.6350	116.1700	116.7049 (39)
												115.8356
HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
HLP (average)	0.8771	0.8752	0.8732	0.8632	0.8612	0.8513	0.8513	0.8493	0.8552	0.8612	0.8652	0.8692 (40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

4. Water heating energy requirements (kWh/year)												
Assumed occupancy												2.9053 (42)
Hot water usage for mixer showers												
	72.9317	71.8357	70.2386	67.1828	64.9277	62.4128	60.9834	62.5684	64.3059	67.0061	70.1275	72.6523 (42a)
Hot water usage for baths												
	31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)
Hot water usage for other uses												
	44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)
Average daily hot water use (litres/day)												136.7815 (43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	148.8009	145.6234	141.7538	135.8699	131.0918	125.9549	123.9809	127.8338	131.8971	137.2900	143.2645	148.4151 (44)
Energy conte	235.6643	207.3657	217.8704	185.9992	176.4747	154.8762	149.9442	158.2849	162.6424	186.3012	204.1065	232.3821 (45)
Energy content (annual)										Total = Sum(45)m =		2271.9117
Distribution loss (46)m = 0.15 x (45)m												
	35.3496	31.1049	32.6806	27.8999	26.4712	23.2314	22.4916	23.7427	24.3964	27.9452	30.6160	34.8573 (46)
Water storage loss (or HIU loss):												
Store volume												150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):												1.9900 (48)
Temperature factor from Table 2b												0.5400 (49)
Enter (49) or (54) in (55)												1.0746 (55)
Total storage loss												
	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (56)
If cylinder contains dedicated solar storage												
	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (57)
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	22.5120	23.2624	22.5120	23.2624 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month												
	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h												
	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (64)
12Total per year (kWh/year)												2938.0367 (64)
Electric shower(s)												2938 (64)
	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)
												0.0000 (64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =												
Heat gains from water heating, kWh/month												
	123.6184	109.8291	117.7019	105.6447	103.9378	95.2963	95.1164	97.8897	97.8786	107.2051	111.6654	122.5271 (65)

5. Internal gains (see Table 5 and 5a)												
Metabolic gains (Table 5), Watts												
(66)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5												
	154.8579	171.4498	154.8579	160.0198	154.8579	160.0198	154.8579	154.8579	160.0198	154.8579	160.0198	154.8579 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5												
	302.7968	305.9389	298.0209	281.1645	259.8864	239.8879	226.5277	223.3857	231.3036	248.1600	269.4381	289.4366 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5												
	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)												
	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110 (71)
Water heating gains (Table 5)												
	166.1537	163.4362	158.2015	146.7288	139.7014	132.3560	127.8447	131.5722	135.9425	144.0929	155.0909	164.6869 (72)
Total internal gains												
	690.3875	707.4039	677.6594	654.4922	621.0248	598.8429	575.8094	576.3949	593.8450	613.6899	651.1279	675.5605 (73)

6. Solar gains							
[Jan]	Area	Solar flux	g	FF	Access	Gains	
	m2	Table 6a	Specific data	Specific data	factor	W	
		W/m2	or Table 6b	or Table 6c	Table 6d		
South	4.4500	46.7521	0.6300	0.7000	0.7700	63.5818 (78)	
North	12.5700	10.6334	0.6300	0.7000	0.7700	40.8488 (74)	
South	7.0000	46.7521	0.6300	0.7000	0.7700	100.0163 (78)	

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West	1.3600		19.6403		0.6300		0.7000		0.7700		8.1632 (80)	
Solar gains	212.6100	361.9644	500.2455	637.1644	735.9945	742.2229	710.6474	634.0027	546.5996	400.8627	254.4902	182.1303 (83)
Total gains	902.9975	1069.3683	1177.9049	1291.6566	1357.0193	1341.0657	1286.4568	1210.3976	1140.4446	1014.5526	905.6181	857.6908 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	45.8477	45.9520	46.0569	46.5883	46.6961	47.2425	47.2425	47.3533	47.0224	46.6961	46.4811	46.2680
alpha	4.0565	4.0635	4.0705	4.1059	4.1131	4.1495	4.1495	4.1569	4.1348	4.1131	4.0987	4.0845
util living area	0.9765	0.9547	0.9183	0.8357	0.7016	0.5245	0.3861	0.4257	0.6392	0.8662	0.9567	0.9805 (86)
Living	19.8719	20.0790	20.3239	20.6104	20.8048	20.8981	20.9194	20.9166	20.8634	20.6095	20.1965	19.8362
Non living	18.8538	19.1154	19.4210	19.7728	19.9929	20.0935	20.1105	20.1105	20.0605	19.7804	19.2728	18.8140
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.4229	20.0790	20.3239	20.6104	20.8048	20.8981	20.9194	20.9166	20.8634	20.6095	20.1965	19.9990 (87)
Th 2	20.1870	20.1887	20.1904	20.1989	20.2006	20.2091	20.2091	20.2108	20.2057	20.2006	20.1972	20.1938 (88)
util rest of house	0.9724	0.9470	0.9046	0.8096	0.6587	0.4661	0.3187	0.3557	0.5803	0.8389	0.9480	0.9770 (89)
MIT 2	19.6555	19.1154	19.4210	19.7728	19.9929	20.0935	20.1105	20.1105	20.0605	19.7804	19.2728	19.0626 (90)
Living area fraction												
MIT	20.0488	19.6093	19.8838	20.2021	20.4091	20.5059	20.5251	20.5237	20.4720	20.2053	19.7462	19.5426 (92)
Temperature adjustment												0.0000
adjusted MIT	20.0488	19.6093	19.8838	20.2021	20.4091	20.5059	20.5251	20.5237	20.4720	20.2053	19.7462	19.5426 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9716	0.9406	0.8988	0.8098	0.6704	0.4884	0.3458	0.3838	0.6007	0.8390	0.9423	0.9736 (94)
Useful gains	877.3305	1005.8314	1058.6527	1045.9298	909.7260	655.0243	444.9208	464.5308	685.0821	851.2507	853.3655	835.0481 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	1854.8147	1728.4505	1569.1140	1309.9425	1007.0724	675.0295	448.6288	470.2207	731.7179	1110.7122	1469.1111	1790.5543 (97)
Space heating kWh	727.2482	485.6001	379.7832	190.0891	72.4257	0.0000	0.0000	0.0000	0.0000	193.0393	443.3369	710.8966 (98a)
Space heating requirement - total per year (kWh/year)												3202.4192
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	727.2482	485.6001	379.7832	190.0891	72.4257	0.0000	0.0000	0.0000	0.0000	193.0393	443.3369	710.8966 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												3202.4192
Space heating per m2										(98c) / (4) =		23.8506 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												365.6366 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												60.0000 (208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating requirement	727.2482	485.6001	379.7832	190.0891	72.4257	0.0000	0.0000	0.0000	0.0000	193.0393	443.3369	710.8966 (98)
Space heating efficiency (main heating system 1)	365.6366	365.6366	365.6366	365.6366	365.6366	0.0000	0.0000	0.0000	0.0000	365.6366	365.6366	365.6366 (210)
Space heating fuel (main heating system)	198.8992	132.8095	103.8690	51.9885	19.8081	0.0000	0.0000	0.0000	0.0000	52.7954	121.2507	194.4271 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating												
Water heating requirement	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (64)
Efficiency of water heater (217)m	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142 (216)
Fuel for water heating, kWh/month	146.9173	129.9383	137.9718	121.0317	117.1610	105.3853	103.8233	108.0164	109.2895	122.1010	130.1348	145.2672 (219)
Space cooling fuel requirement												
(221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	31.6912	28.6243	31.6912	30.6689	31.6912	30.6689	31.6912	31.6912	30.6689	31.6912	30.6689	31.6912 (231)
Lighting	32.1764	25.8131	23.2418	17.0280	13.1529	10.7460	11.9985	15.5961	20.2578	26.5793	30.0213	33.0707 (232)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233a)m	-63.8800	-92.1832	-135.0793	-152.0952	-163.2136	-150.3552	-148.4897	-139.5517	-122.7335	-104.1373	-70.3650	-54.8469 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235c)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233b)m	-26.2757	-58.2607	-121.1440	-191.2945	-261.0154	-266.7295	-263.0830	-220.0414	-158.9000	-88.1853	-36.5727	-20.5981 (233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235d)
Annual totals kWh/year												
Space heating fuel - main system 1												875.8476 (211)
Space heating fuel - main system 2												0.0000 (213)

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Space heating fuel - secondary	0.0000 (215)
Efficiency of water heater	198.9142
Water heating fuel used	1477.0376 (219)
Space cooling fuel	0.0000 (221)
Electricity for pumps and fans: (BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6820)	
mechanical ventilation fans (SFP = 0.6820)	373.1382 (230a)
Total electricity for the above, kWh/year	373.1382 (231)
Electricity for lighting (calculated in Appendix L)	259.6820 (232)
Energy saving/generation technologies (Appendices M ,N and Q)	
PV generation	-3109.0309 (233)
Wind generation	0.0000 (234)
Hydro-electric generation (Appendix N)	0.0000 (235a)
Electricity generated - Micro CHP (Appendix N)	0.0000 (235)
Appendix Q - special features	
Energy saved or generated	-0.0000 (236)
Energy used	0.0000 (237)
Total delivered energy for all uses	-123.3256 (238)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year	
Space heating - main system 1	875.8476	0.1562	136.7917	(261)
Total CO2 associated with community systems			0.0000	(373)
Water heating (other fuel)	1477.0376	0.1409	208.1565	(264)
Space and water heating			344.9483	(265)
Pumps, fans and electric keep-hot	373.1382	0.1387	51.7589	(267)
Energy for lighting	259.6820	0.1443	37.4801	(268)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-1396.9308	0.1346	-187.9788	
PV Unit electricity exported	-1712.1002	0.1245	-213.1868	
Total			-401.1656	(269)
Total CO2, kg/year			33.0217	(272)
EPC Dwelling Carbon Dioxide Emission Rate (DER)			0.2500	(273)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year	
Space heating - main system 1	875.8476	1.5782	1382.2397	(275)
Total CO2 associated with community systems			0.0000	(473)
Water heating (other fuel)	1477.0376	1.5211	2246.7271	(278)
Space and water heating			3628.9668	(279)
Pumps, fans and electric keep-hot	373.1382	1.5128	564.4834	(281)
Energy for lighting	259.6820	1.5338	398.3089	(282)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-1396.9308	1.4973	-2091.6778	
PV Unit electricity exported	-1712.1002	0.4570	-782.4388	
Total			-2874.1166	(283)
Total Primary energy kWh/year			1717.6426	(286)
Dwelling Primary energy Rate (DPER)			12.7900	(287)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022) CALCULATION OF TARGET EMISSIONS

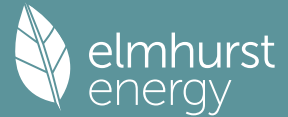
1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)	
Main dwelling	134.2700 (1b)	x 3.3400 (2b)	= 448.4618 (1b) - (3b)	
Ground floor				
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)... (1n)	134.2700			(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)... (3n) =	448.4618 (5)	

2. Ventilation rate

	m3 per hour	
Number of open chimneys	0 * 80 =	0.0000 (6a)
Number of open flues	0 * 20 =	0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 =	0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 =	0.0000 (6d)
Number of flues attached to other heater	0 * 35 =	0.0000 (6e)
Number of blocked chimneys	0 * 20 =	0.0000 (6f)
Number of intermittent extract fans	4 * 10 =	40.0000 (7a)
Number of passive vents	0 * 10 =	0.0000 (7b)
Number of flueless gas fires	0 * 40 =	0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	40.0000 / (5) =	0.0892 (8)
Pressure test		Yes
Pressure Test Method		Blower Door
Measured/design AP50		5.0000 (17)
Infiltration rate		0.3392 (18)
Number of sides sheltered		0 (19)

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Shelter factor
Infiltration rate adjusted to include shelter factor

$$(20) = 1 - [0.075 \times (19)] = 1.0000 \quad (20)$$
$$(21) = (18) \times (20) = 0.3392 \quad (21)$$

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000	(22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750	(22a)
Adj infilt rate													
Effective ac	0.4325	0.4240	0.4155	0.3731	0.3646	0.3222	0.3222	0.3138	0.3392	0.3646	0.3816	0.3986	(22b)
	0.5935	0.5899	0.5863	0.5696	0.5665	0.5519	0.5519	0.5492	0.5575	0.5665	0.5728	0.5794	(25)

3. Heat losses and heat loss parameter

Element	Gross m ²	Openings m ²	NetArea m ²	U-value W/m ² K	A x U W/K	K-value kJ/m ² K	A x K kJ/K	
Main dwelling								
TER Opaque door			1.7100	1.0000	1.7100			(26)
TER Opening Type			25.3800	1.1450	29.0611			(27)
Ground Floor			134.2700	0.1300	17.4551			(28a)
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1800	19.7136			(29a)
External Roof 1 - Pitched	155.0400		155.0400	0.1100	17.0544			(30)
Total net area of external elements Aum(A, m ²)			425.9200					(31)
Fabric heat loss, W/K = Sum (A x U)				(26)...(30) + (32) =	84.9942			(33)
Main dwelling								
Party Wall 1			5.2800	0.0000	0.0000			(32)

Thermal mass parameter (TMP = Cm / TFA) in kJ/m²K

144.7749 (35)

List of Thermal Bridges

	Length	Psi-value	Total	
K1 Element	19.1200	0.0500	0.9560	
E2 Other lintels (including other steel lintels)	10.8000	0.0500	0.5400	
E3 Sill	36.2000	0.0500	1.8100	
E4 Jamb	54.4600	0.1600	8.7136	
E5 Ground floor (normal)	43.4400	0.0400	1.7376	
E11 Eaves (insulation at rafter level)	12.7200	0.0800	1.0176	
E13 Gable (insulation at rafter level)	7.6500	0.0900	0.6885	
E16 Corner (normal)	24.3600	0.0800	1.9488	
R4 Ridge (vaulted ceiling)	5.1000	0.0600	0.3060	
E18 Party wall between dwellings	5.2800	0.0800	0.4224	
P1 Party wall - Ground floor	6.3600	0.0800	0.5088	
P5 Party wall - Roof (insulation at rafter level)				

Thermal bridges (Sum(L x Psi) calculated using Appendix K)

18.6493 (36)

Point Thermal bridges

(36a) = 0.0000

Total fabric heat loss

(33) + (36) + (36a) = 103.6435 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(38)m	87.8359	87.2984	86.7717	84.2975	83.8345	81.6796	81.6796	81.2805	82.5096	83.8345	84.7710	85.7501	(38)
Heat transfer coeff	191.4793	190.9419	190.4151	187.9409	187.4780	185.3230	185.3230	184.9240	186.1531	187.4780	188.4145	189.3935	(39)
Average = Sum(39)m / 12 =													187.9387

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
HLP	1.4261	1.4221	1.4182	1.3997	1.3963	1.3802	1.3802	1.3773	1.3864	1.3963	1.4033	1.4105	(40)
HLP (average)													1.3997
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31	

4. Water heating energy requirements (kWh/year)

Assumed occupancy													2.9053 (42)
Hot water usage for mixer showers													
72.9317	71.8357	70.2386	67.1828	64.9277	62.4128	60.9834	62.5684	64.3059	67.0061	70.1275	72.6523	72.6523	(42a)
Hot water usage for baths													
31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798	31.3798	(42b)
Hot water usage for other uses													
44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829	44.3829	(42c)
Average daily hot water use (litres/day)													136.7815 (43)
Daily hot water use													
148.8009	145.6234	141.7538	135.8699	131.0918	125.9549	123.9809	127.8338	131.8971	137.2900	143.2645	148.4151	148.4151	(44)
Energy conte	235.6643	207.3657	217.8704	185.9992	176.4747	154.8762	149.9442	158.2849	162.6424	186.3012	204.1065	232.3821	(45)
Energy content (annual)													Total = Sum(45)m = 2271.9117
Distribution loss (46)m = 0.15 x (45)m													
35.3496	31.1049	32.6806	27.8999	26.4712	23.2314	22.4916	23.7427	24.3964	27.9452	30.6160	34.8573	34.8573	(46)
Water storage loss (or HIU loss):													
Store volume													150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):													1.3938 (48)
Temperature factor from Table 2b													0.5400 (49)
Enter (49) or (54) in (55)													0.7527 (55)
Total storage loss													
23.3325	21.0745	23.3325	22.5798	23.3325	22.5798	23.3325	23.3325	22.5798	23.3325	22.5798	23.3325	23.3325	(56)
If cylinder contains dedicated solar storage													
23.3325	21.0745	23.3325	22.5798	23.3325	22.5798	23.3325	23.3325	22.5798	23.3325	22.5798	23.3325	23.3325	(57)
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	(59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(61)
Total heat required for water heating calculated for each month													
282.2592	249.4515	264.4653	231.0910	223.0696	199.9681	196.5391	204.8798	207.7342	232.8961	249.1984	278.9770	278.9770	(62)
WWHRS	-33.3414	-29.4874	-30.8775	-25.5678	-23.8282	-20.3900	-19.1124	-20.3241	-21.0963	-24.8702	-28.1749	-32.7239	(63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	(63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(63d)
Output from w/h													
248.9179	219.9641	233.5878	205.5232	199.2414	179.5781	177.4267	184.5557	186.6379	208.0259	221.0235	246.2531	246.2531	(64)
12Total per year (kWh/year)													Total per year (kWh/year) = Sum(64)m = 2510.7352 (64)
Electric shower(s)													2511 (64)
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =													0.0000 (64a)
Heat gains from water heating, kWh/month													
115.6343	102.6177	109.7178	97.9182	95.9538	87.5698	87.1324	89.9056	90.1521	99.2211	103.9389	114.5430	114.5430	(65)

5. Internal gains (see Table 5 and 5a)

6. Solar gains

7. Mean internal temperature (heating season)8. Space heating requirement

9a. Energy requirements - Individual heating systems, including micro-CHP

SAP 10 Online 3.1.57

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Fuel for water heating, kWh/month	285.5541	252.7514	269.2179	238.2861	233.8562	225.0352	222.3392	231.2728	233.8821	241.6090	254.3926	282.3911	(219)
Space cooling fuel requirement													
(221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(221)
Pumps and Fa	7.3041	6.5973	7.3041	7.0685	7.3041	7.0685	7.3041	7.3041	7.0685	7.3041	7.0685	7.3041	(231)
Lighting	32.1764	25.8131	23.2418	17.0280	13.1529	10.7460	11.9985	15.5961	20.2578	26.5793	30.0213	33.0707	(232)
Electricity generated by PVs (Appendix M) (negative quantity)													
(233a)m	-95.7199	-122.1999	-159.2069	-161.6634	-160.8999	-145.4942	-143.4613	-141.4990	-136.9032	-130.4595	-100.4639	-84.3161	(233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)													
(234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)													
(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235c)
Electricity generated by PVs (Appendix M) (negative quantity)													
(233b)m	-100.8778	-204.2165	-392.1582	-570.3806	-737.2784	-734.9588	-726.4590	-622.7364	-466.8533	-285.5998	-132.4081	-80.4171	(233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)													
(234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)													
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235d)
Annual totals kWh/year													
Space heating fuel - main system 1												7335.4205	(211)
Space heating fuel - main system 2												0.0000	(213)
Space heating fuel - secondary												0.0000	(215)
Efficiency of water heater												79.8000	
Water heating fuel used												2970.5878	(219)
Space cooling fuel												0.0000	(221)
Electricity for pumps and fans:													
Total electricity for the above, kWh/year												86.0000	(231)
Electricity for lighting (calculated in Appendix L)												259.6820	(232)
Energy saving/generation technologies (Appendices M ,N and Q)													
PV generation												-6636.6312	(233)
Wind generation												0.0000	(234)
Hydro-electric generation (Appendix N)												0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)												0.0000	(235)
Appendix Q - special features													
Energy saved or generated												-0.0000	(236)
Energy used												0.0000	(237)
Total delivered energy for all uses												4015.0592	(238)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year	
Space heating - main system 1	7335.4205	0.2100	1540.4383	(261)
Total CO2 associated with community systems			0.0000	(373)
Water heating (other fuel)	2970.5878	0.2100	623.8234	(264)
Space and water heating			2164.2617	(265)
Pumps, fans and electric keep-hot	86.0000	0.1387	11.9293	(267)
Energy for lighting	259.6820	0.1443	37.4801	(268)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-1582.2871	0.1368	-216.4593	
PV Unit electricity exported	-5054.3440	0.1269	-641.2181	
Total			-857.6774	(269)
Total CO2, kg/year			1355.9937	(272)
EPC Target Carbon Dioxide Emission Rate (TER)			10.1000	(273)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year	
Space heating - main system 1	7335.4205	1.1300	8289.0252	(275)
Total CO2 associated with community systems			0.0000	(473)
Water heating (other fuel)	2970.5878	1.1300	3356.7642	(278)
Space and water heating			11645.7894	(279)
Pumps, fans and electric keep-hot	86.0000	1.5128	130.1008	(281)
Energy for lighting	259.6820	1.5338	398.3089	(282)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-1582.2871	1.5057	-2382.4725	
PV Unit electricity exported	-5054.3440	0.4657	-2353.9213	
Total			-4736.3938	(283)
Total Primary energy kWh/year			7437.8053	(286)
Target Primary Energy Rate (TPER)			55.3900	(287)

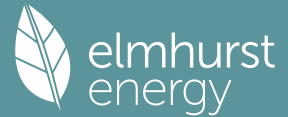
SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)

CALCULATION OF FABRIC ENERGY EFFICIENCY

1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)	
Main dwelling	134.2700	x	448.4618	(1b) - (3b)
Ground floor				
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700			(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	448.4618	(5)

Full SAP Calculation Printout



2. Ventilation rate

												m3 per hour
Number of open chimneys												0 * 80 = 0.0000 (6a)
Number of open flues												0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire												0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler												0 * 20 = 0.0000 (6d)
Number of flues attached to other heater												0 * 35 = 0.0000 (6e)
Number of blocked chimneys												0 * 20 = 0.0000 (6f)
Number of intermittent extract fans												4 * 10 = 40.0000 (7a)
Number of passive vents												0 * 10 = 0.0000 (7b)
Number of flueless gas fires												0 * 40 = 0.0000 (7c)
												Air changes per hour
Infiltration due to chimneys, flues and fans	= (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =											40.0000 / (5) = 0.0892 (8)
Pressure test												Yes
Pressure Test Method												Blower Door
Measured/design AP50												1.0000 (17)
Infiltration rate												0.1392 (18)
Number of sides sheltered												0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] =											1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) =											0.1392 (21)
Wind speed	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind factor	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Adj infilt rate	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
	0.1775	0.1740	0.1705	0.1531	0.1496	0.1322	0.1322	0.1288	0.1392	0.1496	0.1566	0.1636 (22b)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.0000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												0.0000 (23c)
Effective ac	0.5157	0.5151	0.5145	0.5117	0.5112	0.5087	0.5087	0.5083	0.5097	0.5112	0.5123	0.5134 (25)

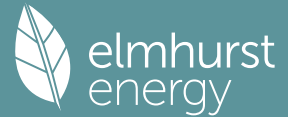
3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K						
Main dwelling													
Door			4.4500	0.9615	4.2788			(27)					
Window			20.9300	0.9615	20.1250			(27)					
Door			1.7100	1.0000	1.7100			(26)					
Ground Floor			134.2700	0.1100	14.7697	110.0000	14769.7000	(28a)					
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1400	15.3328	9.0000	985.6800	(29a)					
External Roof 1 - Pitched	155.0400		155.0400	0.1400	21.7056	9.0000	1395.3600	(30)					
Total net area of external elements Aum(A, m2)			425.9200					(31)					
Fabric heat loss, W/K = Sum (A x U)			(26)...(30) + (32) =		77.9219			(33)					
Main dwelling													
Party Wall 1			5.2800	0.0000	0.0000	20.0000	105.6000	(32)					
Internal Wall 1 - GF			242.5100			9.0000	2182.5900	(32c)					
Heat capacity Cm = Sum(A x k)						(28)...(30) + (32) + (32a)...(32e) =	19438.9300	(34)					
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K							144.7749	(35)					
List of Thermal Bridges													
K1 Element				Length	Psi-value		Total						
E2 Other lintels (including other steel lintels)				19.1200	0.0840		1.6061						
E3 Sill				10.8000	0.0340		0.3672						
E4 Jamb				36.2000	0.0430		1.5566						
E5 Ground floor (normal)				54.4600	0.0210		1.1437						
E11 Eaves (insulation at rafter level)				43.4400	0.0400		1.7376						
E13 Gable (insulation at rafter level)				12.7200	0.0240		0.3053						
E16 Corner (normal)				7.6500	0.0300		0.2295						
R4 Ridge (vaulted ceiling)				24.3600	0.1200		2.9232						
E18 Party wall between dwellings				5.1000	0.0400		0.2040						
P1 Party wall - Ground floor				5.2800	0.3200		1.6896						
P5 Party wall - Roof (insulation at rafter level)				6.3600	0.4800		3.0528						
Thermal bridges (Sum(L x Psi) calculated using Appendix K)							14.8155	(36)					
Point Thermal bridges							(36a) =	0.0000					
Total fabric heat loss						(33) + (36) + (36a) =	92.7375	(37)					
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)													
(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Heat transfer coeff	76.3268	76.2363	76.1476	75.7309	75.6530	75.2901	75.2901	75.2229	75.4299	75.6530	75.8107	75.9756	(38)
Average = Sum(39)m / 12 =	169.0643	168.9738	168.8851	168.4684	168.3904	168.0275	168.0275	167.9603	168.1673	168.3904	168.5482	168.7130	(39)
												168.4680	
HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
HLP (average)	1.2591	1.2585	1.2578	1.2547	1.2541	1.2514	1.2514	1.2509	1.2525	1.2541	1.2553	1.2565	(40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31	

4. Water heating energy requirements (kWh/year)

Assumed occupancy												2.9053 (42)
Hot water usage for mixer showers	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (42a)
Hot water usage for baths	31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)
Hot water usage for other uses	44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)
Average daily hot water use (litres/day)												69.5408 (43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	75.8692	73.7876	71.5152	68.6871	66.1641	63.5420	62.9975	65.2653	67.5912	70.2839	73.1369	75.7627 (44)
Energy content (annual)	120.1582	105.0726	109.9162	94.0293	89.0696	78.1324	76.1900	80.8121	83.3468	95.3745	104.1970	118.6261 (45)
Distribution loss (46)m = 0.15 x (45)m												Total = Sum(45)m = 1154.9248
	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (46)
Water storage loss (or HIU loss):												
Total storage loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (56)

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If cylinder contains dedicated solar storage											
Primary loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (57)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month											
WWHRS	102.1345	89.3117	93.4288	79.9249	75.7091	66.4125	64.7615	68.6903	70.8447	81.0684	88.5674 (62)
PV diverter	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63b)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
Output from w/h	102.1345	89.3117	93.4288	79.9249	75.7091	66.4125	64.7615	68.6903	70.8447	81.0684	88.5674 (64)
Total per year (kWh/year)											982 (64)
Electric shower(s)											981.6861 (64)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =											664.3668 (64a)
Heat gains from water heating, kWh/month											
	40.1352	35.3380	37.5637	33.5383	32.7388	29.7780	29.8044	30.9841	31.2683	34.4736	39.8096 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(66)m	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	154.8579	171.4498	154.8579	160.0198	154.8579	160.0198	154.8579	154.8579	160.0198	154.8579	160.0198	154.8579 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	302.7968	305.9389	298.0209	281.1645	259.8864	239.8879	226.5277	223.3857	231.3036	248.1600	269.4381	289.4366 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110 (71)
Water heating gains (Table 5)	53.9451	52.5864	50.4889	46.5810	44.0037	41.3583	40.0597	41.6453	43.4282	46.3355	50.1128	53.5075 (72)
Total internal gains	578.1789	596.5541	569.9468	554.3445	525.3271	507.8451	488.0244	486.4679	501.3307	515.9325	546.1499	564.3811 (73)

6. Solar gains

[Jan]					Area m2	Solar flux Table 6a W/m2	g Specific data or Table 6b	FF Specific data or Table 6c			Access factor Table 6d	Gains W	
South					4.4500	46.7521	0.6300	0.7000			0.7700	63.5818 (78)	
North					12.5700	10.6334	0.6300	0.7000			0.7700	40.8488 (74)	
South					7.0000	46.7521	0.6300	0.7000			0.7700	100.0163 (78)	
West					1.3600	19.6403	0.6300	0.7000			0.7700	8.1632 (80)	
Solar gains	212.6100	361.9644	500.2455	637.1644	735.9945	742.2229	710.6474	634.0027	546.5996	400.8627	254.4902	182.1303 (83)	
Total gains	790.7888	958.5185	1070.1923	1191.5089	1261.3216	1250.0680	1198.6718	1120.4707	1047.9303	916.7952	800.6401	746.5114 (84)	

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
tau	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
alpha	31.9388	31.9559	31.9726	32.0517	32.0666	32.1358	32.1358	32.1487	32.1091	32.0666	32.0366	32.0053
util living area	3.1293	3.1304	3.1315	3.1368	3.1378	3.1424	3.1424	3.1432	3.1406	3.1378	3.1358	3.1337
	0.9865	0.9749	0.9564	0.9135	0.8332	0.6988	0.5569	0.6048	0.7956	0.9329	0.9770	0.9887 (86)
MIT	18.6260	18.9245	19.3434	19.8949	20.4007	20.7650	20.9165	20.8895	20.6199	19.9575	19.1850	18.5627 (87)
Th 2	19.8730	19.8735	19.8740	19.8765	19.8769	19.8791	19.8791	19.8795	19.8783	19.8769	19.8760	19.8750 (88)
util rest of house	0.9837	0.9697	0.9468	0.8930	0.7905	0.6172	0.4372	0.4870	0.7291	0.9128	0.9713	0.9863 (89)
MIT 2	17.7177	18.0134	18.4266	18.9632	19.4350	19.7446	19.8477	19.8344	19.6366	19.0331	18.2760	17.6563 (90)
Living area fraction	18.1833	18.4804	18.8965	19.4408	19.9300	20.2676	20.3955	20.3752	20.1406	19.5069	18.7419	18.1209 (92)
MIT	18.1833	18.4804	18.8965	19.4408	19.9300	20.2676	20.3955	20.3752	20.1406	19.5069	18.7419	18.1209 (93)
Temperature adjustment												0.0000
adjusted MIT	18.1833	18.4804	18.8965	19.4408	19.9300	20.2676	20.3955	20.3752	20.1406	19.5069	18.7419	18.1209 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9777	0.9611	0.9362	0.8835	0.7927	0.6476	0.4951	0.5423	0.7466	0.9046	0.9635	0.9810 (94)
Useful gains	773.1690	921.2042	1001.9528	1052.7410	999.9094	809.5937	593.4160	607.6331	782.3444	829.3238	771.4158	732.3581 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	2347.1650	2294.7338	2093.5838	1775.7849	1385.8473	952.3117	637.7474	667.6701	1015.8264	1499.8422	1962.2166	2348.6304 (97)
Space heating kWh	1171.0531	923.0119	812.1735	520.5917	287.1378	0.0000	0.0000	0.0000	0.0000	498.8657	857.3766	1202.5066 (98a)
Space heating requirement - total per year (kWh/year)												6272.7167
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	1171.0531	923.0119	812.1735	520.5917	287.1378	0.0000	0.0000	0.0000	0.0000	498.8657	857.3766	1202.5066 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												6272.7167
Space heating per m2												46.7172 (99)

8c. Space cooling requirement

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Calculated for June, July and August. See Table 10b

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Ext. temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000
Heat loss rate W												
Utilisation	0.0000	0.0000	0.0000	0.0000	0.0000	1579.4590	1243.4039	1276.4986	0.0000	0.0000	0.0000	0.0000 (100)
Useful loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.7093	0.7853	0.7502	0.0000	0.0000	0.0000	0.0000 (101)
Total gains	0.0000	0.0000	0.0000	0.0000	0.0000	1120.2933	976.4908	957.6130	0.0000	0.0000	0.0000	0.0000 (102)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	1393.9251	1336.8707	1248.5718	0.0000	0.0000	0.0000	0.0000 (103)
Cooled fraction	0.0000	0.0000	0.0000	0.0000	0.0000	197.0149	268.1226	216.4733	0.0000	0.0000	0.0000	0.0000 (104)
Intermittency factor (Table 10b)	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500 (105)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	49.2537	67.0307	54.1183	0.0000	0.0000	0.0000	0.0000 (107)
Space cooling requirement												170.4027 (107)
Energy for space heating												46.7172 (99)
Energy for space cooling												1.2691 (108)
Total												47.9863 (109)
Fabric Energy Efficiency (DFEE)												48.0 (109)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF TARGET FABRIC ENERGY EFFICIENCY

1. Overall dwelling characteristics

	Area (m ²)	Storey height (m)	Volume (m ³)
Main dwelling	134.2700 (1b)	x 3.3400 (2b)	= 448.4618 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700		(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	448.4618 (5)

2. Ventilation rate

	m ³ per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	4 * 10 = 40.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans	= (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) = 40.0000 / (5) = 0.0892 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	5.0000 (17)
Infiltration rate	0.3392 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.3392 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.4325	0.4240	0.4155	0.3731	0.3646	0.3222	0.3222	0.3138	0.3392	0.3646	0.3816	0.3986 (22b)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.0000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												0.0000 (23c)
Effective ac	0.5935	0.5899	0.5863	0.5696	0.5665	0.5519	0.5519	0.5492	0.5575	0.5665	0.5728	0.5794 (25)

3. Heat losses and heat loss parameter

Element	Gross m ²	Openings m ²	NetArea m ²	U-value W/m ² K	A x U W/K	K-value kJ/m ² K	A x K kJ/K
Main dwelling							
TER Opaque door			1.7100	1.0000	1.7100		(26)
TER Opening Type			25.3800	1.1450	29.0611		(27)
Ground Floor			134.2700	0.1300	17.4551		(28a)
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1800	19.7136		(29a)
External Roof 1 - Pitched	155.0400		155.0400	0.1100	17.0544		(30)
Total net area of external elements Aum(A, m ²)			425.9200				(31)
Fabric heat loss, W/K = Sum (A x U)					(26)...(30) + (32) =	84.9942	(33)
Main dwelling							
Party Wall 1			5.2800	0.0000	0.0000		(32)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m ² K							144.7749 (35)
List of Thermal Bridges							
K1 Element				Length	Psi-value		Total
E2 Other lintels (including other steel lintels)				19.1200	0.0500		0.9560
E3 Sill				10.8000	0.0500		0.5400
E4 Jamb				36.2000	0.0500		1.8100
E5 Ground floor (normal)				54.4600	0.1600		8.7136
E11 Eaves (insulation at rafter level)				43.4400	0.0400		1.7376
E13 Gable (insulation at rafter level)				12.7200	0.0800		1.0176
E16 Corner (normal)				7.6500	0.0900		0.6885
R4 Ridge (vaulted ceiling)				24.3600	0.0800		1.9488

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E18 Party wall between dwellings	5.1000	0.0600	0.3060
P1 Party wall - Ground floor	5.2800	0.0800	0.4224
P5 Party wall - Roof (insulation at rafter level)	6.3600	0.0800	0.5088
Thermal bridges (Sum(L x Psi) calculated using Appendix K)			18.6493 (36)
Point Thermal bridges			(36a) = 0.0000
Total fabric heat loss		(33) + (36) + (36a) =	103.6435 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)												
(38)m	Jan 87.8359	Feb 87.2984	Mar 86.7717	Apr 84.2975	May 83.8345	Jun 81.6796	Jul 81.6796	Aug 81.2805	Sep 82.5096	Oct 83.8345	Nov 84.7710	Dec 85.7501 (38)
Heat transfer coeff												
Average = Sum(39)m / 12 =	191.4793	190.9419	190.4151	187.9409	187.4780	185.3230	185.3230	184.9240	186.1531	187.4780	188.4145	189.3935 (39)
												187.9387
HLP	Jan 1.4261	Feb 1.4221	Mar 1.4182	Apr 1.3997	May 1.3963	Jun 1.3802	Jul 1.3802	Aug 1.3773	Sep 1.3864	Oct 1.3963	Nov 1.4033	Dec 1.4105 (40)
HLP (average)												1.3997
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

4. Water heating energy requirements (kWh/year)

Assumed occupancy												2.9053 (42)
Hot water usage for mixer showers	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (42a)
Hot water usage for baths	31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)
Hot water usage for other uses	44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)
Average daily hot water use (litres/day)												69.5408 (43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Energy content (annual)	75.8692	73.7876	71.5152	68.6871	66.1641	63.5420	62.9975	65.2653	67.5912	70.2839	73.1369	75.7627 (44)
Distribution loss (46)m = 0.15 x (45)m	120.1582	105.0726	109.9162	94.0293	89.0696	78.1324	76.1900	80.8121	83.3468	95.3745	104.1970	118.6261 (45)
Total storage loss	Total = Sum(45)m =											1154.9248
Water storage loss (or HIU loss):	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (46)
If cylinder contains dedicated solar storage	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (56)
Primary loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (57)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (59)
Total heat required for water heating calculated for each month	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
WWHRS	102.1345	89.3117	93.4288	79.9249	75.7091	66.4125	64.7615	68.6903	70.8447	81.0684	88.5674	100.8322 (62)
PV diverter	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63b)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
Output from w/h	102.1345	89.3117	93.4288	79.9249	75.7091	66.4125	64.7615	68.6903	70.8447	81.0684	88.5674	100.8322 (64)
12Total per year (kWh/year)	Total per year (kWh/year) = Sum(64a)m =											981.6861 (64)
Electric shower(s)	58.4062	52.0404	56.8261	54.2285	55.2460	52.6994	54.4560	55.2460	54.2285	56.8261	55.7576	58.4062 (64a)
Heat gains from water heating, kWh/month												664.3668 (64a)
	40.1352	35.3380	37.5637	33.5383	32.7388	29.7780	29.8044	30.9841	31.2683	34.4736	36.0812	39.8096 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts													
(66)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	145.2638	(66)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	154.8579	171.4498	154.8579	160.0198	154.8579	160.0198	154.8579	154.8579	160.0198	154.8579	160.0198	154.8579	(67)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	302.7968	305.9389	298.0209	281.1645	259.8864	239.8879	226.5277	223.3857	231.3036	248.1600	269.4381	289.4366	(68)
Pumps, fans	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	37.5264	(69)
Losses e.g. evaporation (negative values) (Table 5)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(70)
Water heating gains (Table 5)	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	(71)
Total internal gains	53.9451	52.5864	50.4889	46.5810	44.0037	41.3583	40.0597	41.6453	43.4282	46.3355	50.1128	53.5075	(72)
	578.1789	596.5541	569.9468	554.3445	525.3271	507.8451	488.0244	486.4679	501.3307	515.9325	546.1499	564.3811	(73)

6. Solar gains

[Jan]						Area m2	Solar flux Table 6a W/m2	g Specific data or Table 6b	FF Specific data or Table 6c			Access factor Table 6d			Gains W

North						12.5700	10.6334	0.6300	0.7000			0.7700			40.8488 (74)
South						11.4500	46.7521	0.6300	0.7000			0.7700			163.5980 (78)
West						1.3600	19.6403	0.6300	0.7000			0.7700			8.1632 (80)

Solar gains	212.6100	361.9644	500.2455	637.1644	735.9945	742.2229	710.6474	634.0027	546.5996	400.8627	254.4902	182.1303	(83)		
Total gains	790.7888	958.5185	1070.1923	1191.5089	1261.3216	1250.0680	1198.6718	1120.4707	1047.9303	916.7952	800.6401	746.5114	(84)		

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)													21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)													
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		

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tau	28.1999	28.2793	28.3575	28.7309	28.8018	29.1367	29.1367	29.1996	29.0068	28.8018	28.6586	28.5105
alpha	2.8800	2.8853	2.8905	2.9154	2.9201	2.9424	2.9424	2.9466	2.9338	2.9201	2.9106	2.9007
util living area	0.9865	0.9759	0.9594	0.9214	0.8505	0.7264	0.5906	0.6367	0.8160	0.9383	0.9777	0.9885 (86)
MIT	18.3101	18.6179	19.0665	19.6825	20.2480	20.6870	20.8791	20.8454	20.5175	19.7785	18.9451	18.2701 (87)
Th 2	19.7434	19.7465	19.7494	19.7635	19.7662	19.7785	19.7785	19.7808	19.7738	19.7662	19.7608	19.7553 (88)
util rest of house	0.9835	0.9706	0.9499	0.9014	0.8081	0.6411	0.4585	0.5089	0.7487	0.9186	0.9719	0.9860 (89)
MIT 2	17.3186	17.6253	18.0701	18.6796	19.2135	19.6004	19.7341	19.7188	19.4685	18.7837	17.9620	17.2866 (90)
Living area fraction	17.8268	18.1341	18.5808	19.1937	19.7437	20.1574	20.3210	20.2963	20.0062	19.2936	18.4659	17.7907 (91)
MIT	17.8268	18.1341	18.5808	19.1937	19.7437	20.1574	20.3210	20.2963	20.0062	19.2936	18.4659	17.7907 (92)
Temperature adjustment	17.8268	18.1341	18.5808	19.1937	19.7437	20.1574	20.3210	20.2963	20.0062	19.2936	18.4659	17.7907 (93)
adjusted MIT	17.8268	18.1341	18.5808	19.1937	19.7437	20.1574	20.3210	20.2963	20.0062	19.2936	18.4659	17.7907 (93)

8. Space heating requirement

Utilisation	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	0.9770	0.9612	0.9383	0.8900	0.8072	0.6702	0.5211	0.5673	0.7634	0.9089	0.9635	0.9803 (94)
Useful gains	772.5797	921.3696	1004.1656	1060.4662	1018.1062	837.8534	624.6768	635.6338	799.9914	833.2987	771.4097	731.7733 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	2590.0952	2526.9382	2300.3722	1934.5983	1508.0230	1029.9081	689.5798	720.5101	1099.4491	1629.8592	2141.5036	2573.9907 (97)
Space heating kWh	1352.2316	1078.9421	964.3777	629.3751	364.4981	0.0000	0.0000	0.0000	0.0000	592.6410	986.4676	1370.6097 (98a)
Space heating requirement - total per year (kWh/year)												7339.1429
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	1352.2316	1078.9421	964.3777	629.3751	364.4981	0.0000	0.0000	0.0000	0.0000	592.6410	986.4676	1370.6097 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												7339.1429
Space heating per m2										(98c) / (4) =		54.6596 (99)

8c. Space cooling requirement

Calculated for June, July and August. See Table 10b

Ext. temp.	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000
Heat loss rate W	0.0000	0.0000	0.0000	0.0000	0.0000	1742.0366	1371.3905	1405.4222	0.0000	0.0000	0.0000	0.0000 (100)
Utilisation	0.0000	0.0000	0.0000	0.0000	0.0000	0.6583	0.7368	0.7009	0.0000	0.0000	0.0000	0.0000 (101)
Useful loss	0.0000	0.0000	0.0000	0.0000	0.0000	1146.7284	1010.3855	985.0716	0.0000	0.0000	0.0000	0.0000 (102)
Total gains	0.0000	0.0000	0.0000	0.0000	0.0000	1393.9251	1336.8707	1248.5718	0.0000	0.0000	0.0000	0.0000 (103)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	177.9816	242.9050	196.0441	0.0000	0.0000	0.0000	0.0000 (104)
Cooled fraction									fC = cooled area / (4) =			1.0000 (105)
Intermittency factor (Table 10b)	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500 (106)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	44.4954	60.7262	49.0110	0.0000	0.0000	0.0000	0.0000 (107)
Space cooling requirement												154.2327 (107)
Energy for space heating												54.6596 (99)
Energy for space cooling												1.1487 (108)
Total												55.8083 (109)
Fabric Energy Efficiency (TFEE)												55.8 (109)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF ENERGY RATING

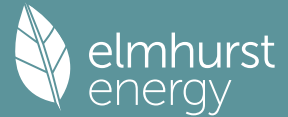
1. Overall dwelling characteristics

Main dwelling	Area (m2)	Storey height (m)	Volume (m3)
Ground floor	134.2700 (1b)	x 3.3400 (2b)	= 448.4618 (1b) - (3b)
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700		(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	448.4618 (5)

2. Ventilation rate

Number of open chimneys	0 * 80 =	0.0000 (6a)
Number of open flues	0 * 20 =	0.0000 (6b)
Number of chimneys / flues attached to closed fire	1 * 10 =	10.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 =	0.0000 (6d)
Number of flues attached to other heater	0 * 35 =	0.0000 (6e)
Number of blocked chimneys	0 * 20 =	0.0000 (6f)
Number of intermittent extract fans	0 * 10 =	0.0000 (7a)
Number of passive vents	0 * 10 =	0.0000 (7b)
Number of flueless gas fires	0 * 40 =	0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	10.0000 / (5) =	0.0223 (8)
Pressure test	Yes	
Pressure Test Method	Blower Door	
Measured/design AP50	1.0000	(17)
Infiltration rate	0.0723	(18)
Number of sides sheltered	0	(19)

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Shelter factor												(20) = 1 - [0.075 x (19)] =	1.0000 (20)
Infiltration rate adjusted to include shelter factor												(21) = (18) x (20) =	0.0723 (21)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000	(22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750	(22a)
Adj infilt rate													
	0.0922	0.0904	0.0886	0.0795	0.0777	0.0687	0.0687	0.0669	0.0723	0.0777	0.0813	0.0850	(22b)
Balanced mechanical ventilation with heat recovery													
If mechanical ventilation													0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)													0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =													84.6000 (23c)
Effective ac	0.1692	0.1674	0.1656	0.1565	0.1547	0.1457	0.1457	0.1439	0.1493	0.1547	0.1583	0.1620	(25)

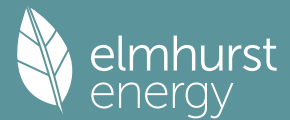
3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K	
Main dwelling								
Door			4.4500	0.9615	4.2788			(27)
Window			20.9300	0.9615	20.1250			(27)
Door			1.7100	1.0000	1.7100			(26)
Ground Floor			134.2700	0.1100	14.7697	110.0000	14769.7000	(28a)
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1400	15.3328	9.0000	985.6800	(29a)
External Roof 1 - Pitched	155.0400		155.0400	0.1400	21.7056	9.0000	1395.3600	(30)
Total net area of external elements Aum(A, m2)			425.9200					(31)
Fabric heat loss, W/K = Sum (A x U)			(26)...(30) + (32) =		77.9219			(33)
Main dwelling								
Party Wall 1			5.2800	0.0000	0.0000	20.0000	105.6000	(32)
Internal Wall 1 - GF			242.5100			9.0000	2182.5900	(32c)
Heat capacity Cm = Sum(A x k)								
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K					(28)...(30) + (32) + (32a)...(32e) =		19438.9300	(34)
List of Thermal Bridges							144.7749	(35)
K1 Element				Length	Psi-value	Total		
E2 Other lintels (including other steel lintels)				19.1200	0.0840	1.6061		
E3 Sill				10.8000	0.0340	0.3672		
E4 Jamb				36.2000	0.0430	1.5566		
E5 Ground floor (normal)				54.4600	0.0210	1.1437		
E11 Eaves (insulation at rafter level)				43.4400	0.0400	1.7376		
E13 Gable (insulation at rafter level)				12.7200	0.0240	0.3053		
E16 Corner (normal)				7.6500	0.0300	0.2295		
R4 Ridge (vaulted ceiling)				24.3600	0.1200	2.9232		
E18 Party wall between dwellings				5.1000	0.0400	0.2040		
P1 Party wall - Ground floor				5.2800	0.3200	1.6896		
P5 Party wall - Roof (insulation at rafter level)				6.3600	0.4800	3.0528		
Thermal bridges (Sum(L x Psi) calculated using Appendix K)							14.8155	(36)
Point Thermal bridges						(36a) =	0.0000	
Total fabric heat loss						(33) + (36) + (36a) =	92.7375	(37)

4. Water heating energy requirements (kWh/year)

Assumed occupancy												2.9053 (42)	
Hot water usage for mixer showers													
72.9317	71.8357	70.2386	67.1828	64.9277	62.4128	60.9834	62.5684	64.3059	67.0061	70.1275	72.6523 (42a)		
Hot water usage for baths													
31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)		
Hot water usage for other uses													
44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)		
Average daily hot water use (litres/day)												136.7815 (43)	
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
148.8009	145.6234	141.7538	135.8699	131.0918	125.9549	123.9809	127.8338	131.8971	137.2900	143.2645	148.4151 (44)		
Energy content (annual)	235.6643	207.3657	217.8704	185.9992	176.4747	154.8762	149.9442	158.2849	162.6424	186.3012	204.1065	232.3821 (45)	
Distribution loss (46)m = 0.15 x (45)m													
35.3496	31.1049	32.6806	27.8999	26.4712	23.2314	22.4916	23.7427	24.3964	27.9452	30.6160	34.8573 (46)		
Water storage loss (or HIU loss):													
Store volume												150.0000 (47)	
a) If manufacturer declared loss factor is known (kWh/day):												1.9900 (48)	
Temperature factor from Table 2b												0.5400 (49)	
Enter (49) or (54) in (55)												1.0746 (55)	
Total storage loss	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (56)	
If cylinder contains dedicated solar storage													
33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (57)		
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	22.5120	23.2624	22.5120	23.2624 (59)	
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)		
Total heat required for water heating calculated for each month	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (62)	
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)	
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)	
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)	
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)	
Output from w/h	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (64)	
Electric shower(s)													
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)		
Total per year (kWh/year) = Sum(64)m =												2938.0367 (64)	

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Heat gains from water heating, kWh/month	Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m = 0.0000 (64a)										
123.6184 109.8291 117.7019 105.6447 103.9378 95.2963 95.1164 97.8897 97.8786 107.2051 111.6654 122.5271 (65)											

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(66)m	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	(66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	36.7607	32.6505	26.5532	20.1025	15.0268	12.6863	13.7080	17.8181	23.9155	30.3662	35.4418	37.7824	(67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	451.9355	456.6252	444.8073	419.6486	387.8901	358.0417	338.1011	333.4114	345.2293	370.3880	402.1465	431.9949	(68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	(69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(70)
Losses e.g. evaporation (negative values) (Table 5)	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	(71)
Water heating gains (Table 5)	166.1537	163.4362	158.2015	146.7288	139.7014	132.3560	127.8447	131.5722	135.9425	144.0929	155.0909	164.6869	(72)
Total internal gains	768.2923	766.1543	743.0044	699.9222	656.0608	616.5264	593.0962	596.2442	618.5296	658.2896	706.1216	747.9066	(73)

6. Solar gains

[Jan]														
	Area				Solar flux		g		FF		Access		Gains	
	m2				Table 6a		Specific data		Specific data		factor		W	
					W/m2		or Table 6b		or Table 6c		Table 6d			
South	4.4500				46.7521		0.6300		0.7000		0.7700		63.5818 (78)	
North	12.5700				10.6334		0.6300		0.7000		0.7700		40.8488 (74)	
South	7.0000				46.7521		0.6300		0.7000		0.7700		100.0163 (78)	
West	1.3600				19.6403		0.6300		0.7000		0.7700		8.1632 (80)	
Solar gains	212.6100	361.9644	500.2455	637.1644	735.9945	742.2229	710.6474	634.0027	546.5996	400.8627	254.4902	182.1303	(83)	
Total gains	980.9023	1128.1187	1243.2499	1337.0866	1392.0553	1358.7493	1303.7436	1230.2470	1165.1292	1059.1523	960.6118	930.0369	(84)	

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)	
Utilisation factor for gains for living area, nil,m (see Table 9a)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
tau	45.8477	45.9520	46.0569	46.5883	46.6961	47.2425	47.2425	47.3533	47.0224	46.6961	46.4811	46.2680	
alpha	4.0565	4.0635	4.0705	4.1059	4.1131	4.1495	4.1495	4.1569	4.1348	4.1131	4.0987	4.0845	
util living area	0.9693	0.9467	0.9055	0.8228	0.6896	0.5185	0.3812	0.4192	0.6288	0.8519	0.9481	0.9744 (86)	
Living	19.9387	20.1247	20.3667	20.6305	20.8124	20.8992	20.9197	20.9171	20.8670	20.6324	20.2397	19.8997	
Non living	18.9378	19.1719	19.4724	19.7955	20.0004	20.0944	20.1106	20.1108	20.0636	19.8059	19.3259	18.8940	
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0	
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0	
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10	
MIT	20.4571	20.1247	20.3667	20.6305	20.8124	20.8992	20.9197	20.9171	20.8670	20.6324	20.2397	20.0536 (87)	
Th 2	20.1870	20.1887	20.1904	20.1989	20.2006	20.2091	20.2091	20.2108	20.2057	20.2006	20.1972	20.1938 (88)	
util rest of house	0.9640	0.9380	0.8902	0.7957	0.6465	0.4606	0.3146	0.3502	0.5701	0.8229	0.9381	0.9700 (89)	
MIT 2	19.6890	19.1719	19.4724	19.7955	20.0004	20.0944	20.1106	20.1108	20.0636	19.8059	19.3259	19.1282 (90)	
Living area fraction	20.0827	19.6603	19.9308	20.2235	20.4166	20.5069	20.5253	20.5241	20.4754	20.2295	19.7943	19.6025 (91)	
MIT	20.0827	19.6603	19.9308	20.2235	20.4166	20.5069	20.5253	20.5241	20.4754	20.2295	19.7943	19.6025 (92)	
Temperature adjustment												0.0000	
adjusted MIT	20.0827	19.6603	19.9308	20.2235	20.4166	20.5069	20.5253	20.5241	20.4754	20.2295	19.7943	19.6025 (93)	

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Utilisation	0.9632	0.9313	0.8848	0.7966	0.6585	0.4828	0.3414	0.3779	0.5907	0.8240	0.9322	0.9661	(94)
Useful gains	944.8270	1050.6668	1100.0310	1065.1095	916.7358	655.9705	445.1209	464.8943	688.2291	872.7558	895.4974	898.5099	(95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000	(96)
Heat loss rate W	1858.8016	1734.4427	1574.6211	1312.4201	1007.9425	675.1436	448.6538	470.2657	732.1035	1113.5092	1474.6927	1797.5503	(97)
Space heating kWh	679.9971	459.4974	353.0950	178.0636	67.8578	0.0000	0.0000	0.0000	0.0000	179.1205	417.0206	668.8861	(98a)
Space heating requirement - total per year (kWh/year)												3003.5382	
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(98b)
Solar heating contribution - total per year (kWh/year)												0.0000	
Space heating kWh	679.9971	459.4974	353.0950	178.0636	67.8578	0.0000	0.0000	0.0000	0.0000	179.1205	417.0206	668.8861	(98c)
Space heating requirement after solar contribution - total per year (kWh/year)												3003.5382	
Space heating per m2										(98c) / (4) =		22.3694	(99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)	
Fraction of space heat from main system(s)												1.0000 (202)	
Efficiency of main space heating system 1 (in %)												365.6366 (206)	
Efficiency of main space heating system 2 (in %)												0.0000 (207)	
Efficiency of secondary/supplementary heating system, %												60.0000 (208)	
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Space heating requirement	679.9971	459.4974	353.0950	178.0636	67.8578	0.0000	0.0000	0.0000	0.0000	179.1205	417.0206	668.8861	(98)
Space heating efficiency (main heating system 1)	365.6366	365.6366	365.6366	365.6366	365.6366	0.0000	0.0000	0.0000	0.0000	365.6366	365.6366	365.6366	(210)

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Space heating fuel (main heating system)	185.9762	125.6705	96.5699	48.6996	18.5588	0.0000	0.0000	0.0000	0.0000	48.9887	114.0533	182.9374	(211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(215)
Water heating													
Water heating requirement	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571	(64)
Efficiency of water heater													(216)
(217)m	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	(217)
Fuel for water heating, kWh/month	146.9173	129.9383	137.9718	121.0317	117.1610	105.3853	103.8233	108.0164	109.2895	122.1010	130.1348	145.2672	(219)
Space cooling fuel requirement													
(221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(221)
Pumps and Fa	31.6912	28.6243	31.6912	30.6689	31.6912	30.6689	31.6912	31.6912	30.6689	31.6912	30.6689	31.6912	(231)
Lighting	32.1764	25.8131	23.2418	17.0280	13.1529	10.7460	11.9985	15.5961	20.2578	26.5793	30.0213	33.0707	(232)
Electricity generated by PVs (Appendix M) (negative quantity)													
(233a)m	-63.6878	-91.9244	-134.5769	-151.7091	-163.0098	-150.3552	-148.4897	-139.5517	-122.7335	-103.9006	-70.1889	-54.7075	(233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)													
(234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)													
(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235c)
Electricity generated by PVs (Appendix M) (negative quantity)													
(233b)m	-26.4679	-58.5195	-121.6463	-191.6806	-261.2192	-266.7295	-263.0830	-220.0414	-158.9000	-88.4220	-36.7488	-20.7374	(233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)													
(234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)													
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235d)
Annual totals kWh/year													
Space heating fuel - main system 1												821.4545	(211)
Space heating fuel - main system 2												0.0000	(213)
Space heating fuel - secondary												0.0000	(215)
Efficiency of water heater												198.9142	
Water heating fuel used												1477.0376	(219)
Space cooling fuel												0.0000	(221)

Electricity for pumps and fans:													
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6820)													
mechanical ventilation fans (SFP = 0.6820)												373.1382	(230a)
Total electricity for the above, kWh/year												373.1382	(231)
Electricity for lighting (calculated in Appendix L)												259.6820	(232)

Energy saving/generation technologies (Appendices M ,N and Q)													
PV generation												-3109.0309	(233)
Wind generation												0.0000	(234)
Hydro-electric generation (Appendix N)												0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)												0.0000	(235)
Appendix Q - special features													
Energy saved or generated												-0.0000	(236)
Energy used												0.0000	(237)
Total delivered energy for all uses												-177.7187	(238)

10a. Fuel costs - using Table 12 prices

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year
Space heating - main system 1	821.4545	16.4900	135.4578 (240)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1477.0376	16.4900	243.5635 (247)
Energy for instantaneous electric shower(s)	0.0000	16.4900	0.0000 (247a)
Pumps, fans and electric keep-hot	373.1382	16.4900	61.5305 (249)
Energy for lighting	259.6820	16.4900	42.8216 (250)
Additional standing charges			0.0000 (251)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1394.8352	16.4900	-230.0083
PV Unit electricity exported	-1714.1957	5.5900	-95.8235
Total			-325.8319 (252)
Total energy cost			157.5415 (255)

11a. SAP rating - Individual heating systems

Energy cost deflator (Table 12):		0.3600 (256)
Energy cost factor (ECF)		0.3164 (257)
SAP value		94.8717
SAP rating (Section 12)		95 (258)
SAP band		A

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	821.4545	0.1562	128.3137 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	1477.0376	0.1409	208.1565 (264)
Space and water heating			336.4703 (265)
Pumps, fans and electric keep-hot	373.1382	0.1387	51.7589 (267)
Energy for lighting	259.6820	0.1443	37.4801 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1394.8352	0.1345	-187.6656
PV Unit electricity exported	-1714.1957	0.1246	-213.5368
Total			-401.2024 (269)
Total CO2, kg/year			24.5069 (272)

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CO2 emissions per m2 0.1800 (273)
 EI value 99.8168
 EI rating 100 (274)
 EI band A

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
 CALCULATION OF EPC COSTS, EMISSIONS AND PRIMARY ENERGY

1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)
Main dwelling	134.2700 (1b)	x 3.3400 (2b)	= 448.4618 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700		(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n)	= 448.4618 (5)

2. Ventilation rate

m3 per hour

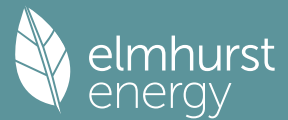
Number of open chimneys	0 * 80 =	0.0000 (6a)
Number of open flues	0 * 20 =	0.0000 (6b)
Number of chimneys / flues attached to closed fire	1 * 10 =	10.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 =	0.0000 (6d)
Number of flues attached to other heater	0 * 35 =	0.0000 (6e)
Number of blocked chimneys	0 * 20 =	0.0000 (6f)
Number of intermittent extract fans	0 * 10 =	0.0000 (7a)
Number of passive vents	0 * 10 =	0.0000 (7b)
Number of flueless gas fires	0 * 40 =	0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	10.0000 / (5) =	0.0223 (8)
Pressure test	Yes	
Pressure Test Method	Blower Door	
Measured/design AP50		1.0000 (17)
Infiltration rate		0.0723 (18)
Number of sides sheltered		0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] =	1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) =	0.0723 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.7000	5.4000	5.4000	4.8000	4.8000	4.2000	4.2000	4.1000	4.4000	5.0000	5.1000	5.6000 (22)
Wind factor	1.4250	1.3500	1.3500	1.2000	1.2000	1.0500	1.0500	1.0250	1.1000	1.2500	1.2750	1.4000 (22a)
Adj infilt rate	0.1030	0.0976	0.0976	0.0868	0.0868	0.0759	0.0759	0.0741	0.0795	0.0904	0.0922	0.1012 (22b)
Balanced mechanical ventilation with heat recovery												
If mechanical ventilation												0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												84.6000 (23c)
Effective ac	0.1800	0.1746	0.1746	0.1638	0.1638	0.1529	0.1529	0.1511	0.1565	0.1674	0.1692	0.1782 (25)

3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K					
Main dwelling												
Door			4.4500	0.9615	4.2788		(27)					
Window			20.9300	0.9615	20.1250		(27)					
Door			1.7100	1.0000	1.7100		(26)					
Ground Floor			134.2700	0.1100	14.7697	110.0000	14769.7000 (28a)					
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1400	15.3328	9.0000	985.6800 (29a)					
External Roof 1 - Pitched	155.0400		155.0400	0.1400	21.7056	9.0000	1395.3600 (30)					
Total net area of external elements Aum(A, m2)			425.9200				(31)					
Fabric heat loss, W/K = Sum (A x U)			(26)...(30) + (32) =		77.9219		(33)					
Main dwelling												
Party Wall 1			5.2800	0.0000	0.0000	20.0000	105.6000 (32)					
Internal Wall 1 - GF			242.5100			9.0000	2182.5900 (32c)					
Heat capacity Cm = Sum(A x k)					(28)...(30) + (32) + (32a)...(32e) =		19438.9300 (34)					
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K							144.7749 (35)					
List of Thermal Bridges												
K1 Element				Length	Psi-value	Total						
E2 Other lintels (including other steel lintels)				19.1200	0.0840	1.6061						
E3 Sill				10.8000	0.0340	0.3672						
E4 Jamb				36.2000	0.0430	1.5566						
E5 Ground floor (normal)				54.4600	0.0210	1.1437						
E11 Eaves (insulation at rafter level)				43.4400	0.0400	1.7376						
E13 Gable (insulation at rafter level)				12.7200	0.0240	0.3053						
E16 Corner (normal)				7.6500	0.0300	0.2295						
R4 Ridge (vaulted ceiling)				24.3600	0.1200	2.9232						
E18 Party wall between dwellings				5.1000	0.0400	0.2040						
P1 Party wall - Ground floor				5.2800	0.3200	1.6896						
P5 Party wall - Roof (insulation at rafter level)				6.3600	0.4800	3.0528						
Thermal bridges (Sum(L x Psi) calculated using Appendix K)							14.8155 (36)					
Point Thermal bridges						(36a) =	0.0000					
Total fabric heat loss						(33) + (36) + (36a) =	92.7375 (37)					
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)												
(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	26.6424	25.8399	25.8399	24.2350	24.2350	22.6300	22.6300	22.3625	23.1650	24.7699	25.0374	26.3749 (38)

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Heat transfer coeff	119.3798	118.5774	118.5774	116.9724	116.9724	115.3675	115.3675	115.1000	115.9025	117.5074	117.7749	119.1123 (39)
Average = Sum(39)m / 12 =												117.2176
HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
HLP (average)	0.8891	0.8831	0.8831	0.8712	0.8712	0.8592	0.8592	0.8572	0.8632	0.8752	0.8771	0.8871 (40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

4. Water heating energy requirements (kWh/year)

Assumed occupancy												2.9053 (42)
Hot water usage for mixer showers	72.9317	71.8357	70.2386	67.1828	64.9277	62.4128	60.9834	62.5684	64.3059	67.0061	70.1275	72.6523 (42a)
Hot water usage for baths	31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)
Hot water usage for other uses	44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)
Average daily hot water use (litres/day)												136.7815 (43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Energy conte	148.8009	145.6234	141.7538	135.8699	131.0918	125.9549	123.9809	127.8338	131.8971	137.2900	143.2645	148.4151 (44)
Energy content (annual)	235.6643	207.3657	217.8704	185.9992	176.4747	154.8762	149.9442	158.2849	162.6424	186.3012	204.1065	232.3821 (45)
Distribution loss (46)m = 0.15 x (45)m												2271.9117
Water storage loss (or HIU loss):	35.3496	31.1049	32.6806	27.8999	26.4712	23.2314	22.4916	23.7427	24.3964	27.9452	30.6160	34.8573 (46)
Store volume												150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):												1.9900 (48)
Temperature factor from Table 2b												0.5400 (49)
Enter (49) or (54) in (55)												1.0746 (55)
Total storage loss	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (56)
If cylinder contains dedicated solar storage	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (57)
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	22.5120	23.2624	22.5120	23.2624 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571 (64)
Electric shower(s)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =												0.0000 (64a)
Heat gains from water heating, kWh/month	123.6184	109.8291	117.7019	105.6447	103.9378	95.2963	95.1164	97.8897	97.8786	107.2051	111.6654	122.5271 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(66)m	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	36.7607	32.6505	26.5532	20.1025	15.0268	12.6863	13.7080	17.8181	23.9155	30.3662	35.4418	37.7824 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	451.9355	456.6252	444.8073	419.6486	387.8901	358.0417	338.1011	333.4114	345.2293	370.3880	402.1465	431.9949 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110 (71)
Water heating gains (Table 5)	166.1537	163.4362	158.2015	146.7288	139.7014	132.3560	127.8447	131.5722	135.9425	144.0929	155.0909	164.6869 (72)
Total internal gains	768.2923	766.1543	743.0044	699.9222	656.0608	616.5264	593.0962	596.2442	618.5296	658.2896	706.1216	747.9066 (73)

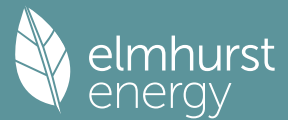
6. Solar gains

[Jan]												
					Area m2	Solar flux Table 6a W/m2		g Specific data or Table 6b	FF Specific data or Table 6c	Access factor Table 6d		Gains W
South					4.4500	58.8467		0.6300	0.7000	0.7700		80.0301 (78)
North					12.5700	14.1647		0.6300	0.7000	0.7700		54.4144 (74)
South					7.0000	58.8467		0.6300	0.7000	0.7700		125.8901 (78)
West					1.3600	26.5321		0.6300	0.7000	0.7700		11.0276 (80)
Solar gains	271.3623	397.0642	542.6195	712.6389	785.0878	867.4216	758.4583	721.7457	629.3924	457.1585	313.0134	223.7643 (83)
Total gains	1039.6546	1163.2185	1285.6239	1412.5611	1441.1486	1483.9480	1351.5544	1317.9899	1247.9220	1115.4481	1019.1350	971.6709 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	45.2313	45.5374	45.5374	46.1622	46.1622	46.8044	46.8044	46.9131	46.5883	45.9520	45.8477	45.3329
alpha	4.0154	4.0358	4.0358	4.0775	4.0775	4.1203	4.1203	4.1275	4.1059	4.0635	4.0565	4.0222
util living area	0.9452	0.9193	0.8716	0.7866	0.6724	0.4966	0.4115	0.4043	0.5711	0.7852	0.9043	0.9512 (86)
Living	20.1886	20.3173	20.4931	20.6855	20.8202	20.9011	20.9161	20.9172	20.8852	20.7363	20.4699	20.1816

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Non living	19.2445	19.4063	19.6187	19.8517	19.9998	20.0885	20.1010	20.1036	20.0727	19.9127	19.6015	19.2384
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.5849	20.3173	20.4931	20.6855	20.8202	20.9011	20.9161	20.9172	20.8852	20.7363	20.4699	20.2961 (87)
Th 2	20.1768	20.1819	20.1819	20.1921	20.1921	20.2023	20.2023	20.2040	20.1989	20.1887	20.1870	20.1785 (88)
util rest of house												
	0.9352	0.9055	0.8508	0.7555	0.6282	0.4416	0.3471	0.3383	0.5103	0.7447	0.8850	0.9418 (89)
MIT 2	19.8051	19.4063	19.6187	19.8517	19.9998	20.0885	20.1010	20.1036	20.0727	19.9127	19.6015	19.4078 (90)
Living area fraction												0.5125 (91)
MIT	20.2048	19.8732	20.0669	20.2791	20.4203	20.5050	20.5188	20.5206	20.4891	20.3349	20.0466	19.8631 (92)
Temperature adjustment												0.0000
adjusted MIT	20.2048	19.8732	20.0669	20.2791	20.4203	20.5050	20.5188	20.5206	20.4891	20.3349	20.0466	19.8631 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Utilisation	0.9356	0.8995	0.8476	0.7591	0.6411	0.4628	0.3729	0.3648	0.5329	0.7525	0.8812	0.9376	(94)
Useful gains	972.6518	1046.3688	1089.6752	1072.2264	923.9848	686.7664	504.0281	480.7771	664.9811	839.3660	898.0922	911.0603	(95)
Ext temp.	6.4000	6.7000	7.8000	9.4000	11.8000	14.4000	16.1000	16.3000	14.5000	11.9000	9.2000	6.8000	(96)
Heat loss rate W	1648.0133	1562.0480	1454.5728	1272.5498	1008.3367	704.3208	509.7821	485.7917	694.1551	991.1591	1277.4583	1555.9740	(97)
Space heating kWh	502.4690	346.5364	271.4838	144.2328	62.7578	0.0000	0.0000	0.0000	0.0000	112.9340	273.1436	479.8158	(98a)
Space heating requirement - total per year (kWh/year)												2193.3732	
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(98b)
Solar heating contribution - total per year (kWh/year)												0.0000	
Space heating kWh	502.4690	346.5364	271.4838	144.2328	62.7578	0.0000	0.0000	0.0000	0.0000	112.9340	273.1436	479.8158	(98c)
Space heating requirement after solar contribution - total per year (kWh/year)												2193.3732	
Space heating per m2										(98c) / (4) =		16.3355	(99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)													0.0000	(201)
Fraction of space heat from main system(s)													1.0000	(202)
Efficiency of main space heating system 1 (in %)													364.9749	(206)
Efficiency of main space heating system 2 (in %)													0.0000	(207)
Efficiency of secondary/supplementary heating system, %													60.0000	(208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
Space heating requirement	502.4690	346.5364	271.4838	144.2328	62.7578	0.0000	0.0000	0.0000	0.0000	112.9340	273.1436	479.8158	(98)	
Space heating efficiency (main heating system 1)	364.9749	364.9749	364.9749	364.9749	364.9749	0.0000	0.0000	0.0000	0.0000	364.9749	364.9749	364.9749	(210)	
Space heating fuel (main heating system)	137.6722	94.9480	74.3842	39.5185	17.1951	0.0000	0.0000	0.0000	0.0000	30.9430	74.8390	131.4654	(211)	
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(212)	
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(213)	
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(215)	
Water heating														
Water heating requirement	292.2393	258.4657	274.4454	240.7492	233.0497	209.6262	206.5192	214.8599	217.3924	242.8762	258.8565	288.9571	(64)	
Efficiency of water heater (217)m	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	(216)	
Fuel for water heating, kWh/month	147.0040	130.0150	138.0532	121.1031	117.2301	105.4474	103.8845	108.0801	109.3540	122.1730	130.2116	145.3530	(219)	
Space cooling fuel requirement (221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(221)	
Pumps and Fa	31.6912	28.6243	31.6912	30.6689	31.6912	30.6689	31.6912	31.6912	30.6689	31.6912	30.6689	31.6912	(231)	
Lighting	32.1764	25.8131	23.2418	17.0280	13.1529	10.7460	11.9985	15.5961	20.2578	26.5793	30.0213	33.0707	(232)	
Electricity generated by PVs (Appendix M) (negative quantity) (233a)m	-78.5991	-99.5020	-142.8012	-161.8607	-168.6632	-161.6466	-153.3933	-149.7057	-134.4744	-114.5172	-83.1007	-65.7011	(233a)	
Electricity generated by wind turbines (Appendix M) (negative quantity) (234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234a)	
Electricity generated by hydro-electric generators (Appendix M) (negative quantity) (235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235a)	
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation) (235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235c)	
Electricity generated by PVs (Appendix M) (negative quantity) (233b)m	-40.3329	-71.0144	-142.7337	-228.1639	-286.5772	-326.9661	-287.5775	-264.2297	-197.2860	-111.7933	-52.8689	-30.0657	(233b)	
Electricity generated by wind turbines (Appendix M) (negative quantity) (234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234b)	
Electricity generated by hydro-electric generators (Appendix M) (negative quantity) (235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235b)	
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation) (235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235d)	
Annual totals kWh/year														
Space heating fuel - main system 1													600.9654	(211)
Space heating fuel - main system 2													0.0000	(213)
Space heating fuel - secondary													0.0000	(215)
Efficiency of water heater													198.7969	
Water heating fuel used													1477.9090	(219)
Space cooling fuel													0.0000	(221)
Electricity for pumps and fans:														
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6820)														
mechanical ventilation fans (SFP = 0.6820)													373.1382	(230a)
Total electricity for the above, kWh/year													373.1382	(231)
Electricity for lighting (calculated in Appendix L)													259.6820	(232)
Energy saving/generation technologies (Appendices M ,N and Q)														
PV generation													-3553.5745	(233)
Wind generation													0.0000	(234)
Hydro-electric generation (Appendix N)													0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)													0.0000	(235)
Appendix Q - special features														
Energy saved or generated													-0.0000	(236)
Energy used													0.0000	(237)
Total delivered energy for all uses													-841.8799	(238)

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10a. Fuel costs - using BEDF prices (595)

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year
Space heating - main system 1	600.9654	27.6700	166.2871 (240)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1477.9090	27.6700	408.9374 (247)
Energy for instantaneous electric shower(s)	0.0000	27.6700	0.0000 (247a)
Pumps, fans and electric keep-hot	373.1382	27.6700	103.2473 (249)
Energy for lighting	259.6820	27.6700	71.8540 (250)
Additional standing charges			0.0000 (251)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1513.9652	27.6700	-418.9142
PV Unit electricity exported	-2039.6093	5.8100	-118.5013
Total			-537.4155 (252)
Total energy cost			212.9104 (255)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	600.9654	0.1562	93.8717 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	1477.9090	0.1409	208.2793 (264)
Space and water heating			302.1511 (265)
Pumps, fans and electric keep-hot	373.1382	0.1387	51.7589 (267)
Energy for lighting	259.6820	0.1443	37.4801 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1513.9652	0.1350	-204.4192
PV Unit electricity exported	-2039.6093	0.1255	-255.9295
Total			-460.3487 (269)
Total CO2, kg/year			-68.9586 (272)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year
Space heating - main system 1	600.9654	1.5783	948.4779 (275)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1477.9090	1.5211	2248.0526 (278)
Space and water heating			3196.5305 (279)
Pumps, fans and electric keep-hot	373.1382	1.5128	564.4834 (281)
Energy for lighting	259.6820	1.5338	398.3089 (282)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1513.9652	1.4990	-2269.4780
PV Unit electricity exported	-2039.6093	0.4606	-939.3980
Total			-3208.8759 (283)
Total Primary energy kWh/year			950.4469 (286)

SAP 10 EPC IMPROVEMENTS

Plot 1

Current energy efficiency rating: A 95
Current environmental impact rating: A 100

N Solar water heating			Recommended
U Solar photovoltaic panels			Already installed
V2 Wind turbine			Recommended
Recommended measures:	SAP change	Cost change	CO2 change
N Solar water heating	+ 1.3	-£ 81	-42 kg (61.1%)
V2 Wind turbine	+ 15.4	-£ 755	-496 kg (446.5%)

Recommended measures	Typical annual savings		Energy efficiency	Environmental impact
Solar water heating	£81	0.31 kg/m²	A 96	A 100
Wind turbine	£755	3.69 kg/m²	A 112	A 104
Total Savings	£835	4.01 kg/m²		

Potential energy efficiency rating: A 112
Potential environmental impact rating: A 104

Fuel prices for cost data on this page from database revision number 595 TEST (29 May 2026)
Recommendation texts revision number 6.1 (11 Jun 2019)

Typical heating and lighting costs of this home (per year, South West England):

	Current £750	Potential £656	Saving £94
Electricity			
Space heating	£270	£292	-£22
Water heating	£409	£293	£116
Lighting	£72	£72	£0
Generated (PV)	-£537	-£524	-£13
Generated (wind)	-£0	-£755	£755
Total cost of fuels	£213	-£623	£836
Total cost of uses	£214	-£622	£836
Delivered energy	-6 kWh/m²	-35 kWh/m²	29 kWh/m²

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Carbon dioxide emissions	-0.1 tonnes	-0.6 tonnes	0.5 tonnes
CO2 emissions per m ²	-1 kg/m ²	-5 kg/m ²	4 kg/m ²
Primary energy	7 kWh/m ²	-28 kWh/m ²	36 kWh/m ²

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF ENERGY RATING FOR IMPROVED DWELLING

1. Overall dwelling characteristics

	Area (m ²)	Storey height (m)	Volume (m ³)
Main dwelling	134.2700 (1b)	x 3.3400 (2b)	= 448.4618 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) = 448.4618 (5)
Dwelling volume			

2. Ventilation rate

	m ³ per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	1 * 10 = 10.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	0 * 10 = 0.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	10.0000 / (5) = 0.0223 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	1.0000 (17)
Infiltration rate	0.0723 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.0723 (21)

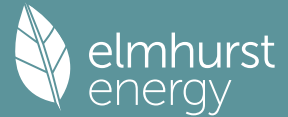
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.0922	0.0904	0.0886	0.0795	0.0777	0.0687	0.0687	0.0669	0.0723	0.0777	0.0813	0.0850 (22b)
Balanced mechanical ventilation with heat recovery												
If mechanical ventilation												0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												84.6000 (23c)
Effective ac	0.1692	0.1674	0.1656	0.1565	0.1547	0.1457	0.1457	0.1439	0.1493	0.1547	0.1583	0.1620 (25)

3. Heat losses and heat loss parameter

Element	Gross m ²	Openings m ²	NetArea m ²	U-value W/m ² K	A x U W/K	K-value kJ/m ² K	A x K kJ/K
Main dwelling							
Door			4.4500	0.9615	4.2788		(27)
Window			20.9300	0.9615	20.1250		(27)
Door			1.7100	1.0000	1.7100		(26)
Ground Floor			134.2700	0.1100	14.7697	110.0000	14769.7000 (28a)
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1400	15.3328	9.0000	985.6800 (29a)
External Roof 1 - Pitched	155.0400		155.0400	0.1400	21.7056	9.0000	1395.3600 (30)
Total net area of external elements Aum(A, m ²)			425.9200				(31)
Fabric heat loss, W/K = Sum (A x U)				(26)...(30) + (32) =	77.9219		(33)
Main dwelling							
Party Wall 1			5.2800	0.0000	0.0000	20.0000	105.6000 (32)
Internal Wall 1 - GF			242.5100			9.0000	2182.5900 (32c)
Heat capacity Cm = Sum(A x k)					(28)...(30) + (32) + (32a)...(32e) =		19438.9300 (34)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m ² K							144.7749 (35)
List of Thermal Bridges							
K1 Element				Length	Psi-value	Total	
E2 Other lintels (including other steel lintels)				19.1200	0.0840	1.6061	
E3 Sill				10.8000	0.0340	0.3672	
E4 Jamb				36.2000	0.0430	1.5566	
E5 Ground floor (normal)				54.4600	0.0210	1.1437	
E11 Eaves (insulation at rafter level)				43.4400	0.0400	1.7376	
E13 Gable (insulation at rafter level)				12.7200	0.0240	0.3053	
E16 Corner (normal)				7.6500	0.0300	0.2295	
R4 Ridge (vaulted ceiling)				24.3600	0.1200	2.9232	
E18 Party wall between dwellings				5.1000	0.0400	0.2040	
P1 Party wall - Ground floor				5.2800	0.3200	1.6896	
P5 Party wall - Roof (insulation at rafter level)				6.3600	0.4800	3.0528	
Thermal bridges (Sum(L x Psi) calculated using Appendix K)							14.8155 (36)
Point Thermal bridges						(36a) =	0.0000
Total fabric heat loss						(33) + (36) + (36a) =	92.7375 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)												
(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	25.0374	24.7699	24.5024	23.1650	22.8975	21.5601	21.5601	21.2926	22.0950	22.8975	23.4325	23.9675 (38)

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Heat transfer coeff	117.7749	117.5074	117.2399	115.9025	115.6350	114.2975	114.2975	114.0300	114.8325	115.6350	116.1700	116.7049 (39)
Average = Sum(39)m / 12 =												115.8356
HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
HLP (average)	0.8771	0.8752	0.8732	0.8632	0.8612	0.8513	0.8513	0.8493	0.8552	0.8612	0.8652	0.8692 (40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

4. Water heating energy requirements (kWh/year)

Assumed occupancy													2.9053 (42)
Hot water usage for mixer showers													
72.9317	71.8357	70.2386	67.1828	64.9277	62.4128	60.9834	62.5684	64.3059	67.0061	70.1275	72.6523 (42a)		
Hot water usage for baths													
31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)		
Hot water usage for other uses													
44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)		
Average daily hot water use (litres/day)													136.7815 (43)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Daily hot water use	148.8009	145.6234	141.7538	135.8699	131.0918	125.9549	123.9809	127.8338	131.8971	137.2900	143.2645	148.4151 (44)	
Energy content	235.6643	207.3657	217.8704	185.9992	176.4747	154.8762	149.9442	158.2849	162.6424	186.3012	204.1065	232.3821 (45)	
Energy content (annual)										Total = Sum(45)m =		2271.9117	
Distribution loss (46)m = 0.15 x (45)m													
35.3496	31.1049	32.6806	27.8999	26.4712	23.2314	22.4916	23.7427	24.3964	27.9452	30.6160	34.8573 (46)		
Water storage loss (or HIU loss):													
Store volume													150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):													1.9900 (48)
Temperature factor from Table 2b													0.5400 (49)
Enter (49) or (54) in (55)													1.0746 (55)
Total storage loss													
33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (56)		
If cylinder contains dedicated solar storage													
33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (57)		
Primary loss	23.2624	21.0112	21.8667	15.7584	10.4681	9.9053	10.2355	11.1660	17.1091	21.8667	22.5120	23.2624 (59)	
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)		
Total heat required for water heating calculated for each month													
292.2393	258.4657	273.0496	233.9956	220.2554	197.0195	193.4922	202.7634	211.9895	241.4804	258.8565	288.9571 (62)		
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)		
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)		
Aperture area of solar collector													3.0000 (H1)
Zero-loss collector efficiency													0.8000 (H2)
Collector linear heat loss coefficient													1.8000 (H3)
Collector 2nd order heat loss coefficient													0.0000 (H4)
Collector loop efficiency													0.9000 (H5)
Incidence angle modifier													1.0000 (H6)
Overshading factor													0.8000 (H8)
Overall heat loss coefficient of system													6.5000 (H10)
Heat loss coefficient of collector loop													3.9667 (H11)
Dedicated solar storage volume													75.0000 (H12)
Effective solar volume													75.0000 (H14)
Reference volume													225.0000 (H15)
Storage tank correction coefficient													1.3161 (H16)
Heat delivered to hot water													632.8165 (H24)
Heat delivered to space heating													0.0000 (H29)
Solar input													632.8165
Solar input	-0.0000	-16.1928	-58.8748	-81.3965	-106.9565	-98.7079	-98.0576	-85.3774	-58.5005	-28.7525	-0.0000	-0.0000 (63c)	
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)		
Output from w/h													
292.2393	242.2730	214.1748	152.5991	113.2989	98.3116	95.4346	117.3860	153.4890	212.7280	258.8565	288.9571 (64)		
Electric shower(s)													2239.7479 (64)
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)		
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =													0.0000 (64a)
Heat gains from water heating, kWh/month													
123.6184	109.8291	116.5853	100.2418	93.7024	85.2110	84.6949	88.2126	93.5563	106.0886	111.6654	122.5271 (65)		

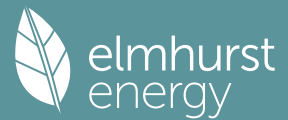
5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(66)m	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	36.7607	32.6505	26.5532	20.1025	15.0268	12.6863	13.7080	17.8181	23.9155	30.3662	35.4418	37.7824 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	451.9355	456.6252	444.8073	419.6486	387.8901	358.0417	338.1011	333.4114	345.2293	370.3880	402.1465	431.9949 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110 (71)
Water heating gains (Table 5)	166.1537	163.4362	156.7007	139.2248	125.9441	118.3486	113.8372	118.5653	129.9393	142.5921	155.0909	164.6869 (72)
Total internal gains	768.2923	766.1543	741.5036	692.4182	642.3035	602.5190	579.0887	583.2373	612.5264	656.7888	706.1216	747.9066 (73)

6. Solar gains

[Jan]			Area m2	Solar flux Table 6a W/m2	g Specific data or Table 6b	FF Specific data or Table 6c	Access factor Table 6d	Gains W				
South			4.4500	46.7521	0.6300	0.7000	0.7700	63.5818 (78)				
North			12.5700	10.6334	0.6300	0.7000	0.7700	40.8488 (74)				
South			7.0000	46.7521	0.6300	0.7000	0.7700	100.0163 (78)				
West			1.3600	19.6403	0.6300	0.7000	0.7700	8.1632 (80)				
Solar gains	212.6100	361.9644	500.2455	637.1644	735.9945	742.2229	710.6474	634.0027	546.5996	400.8627	254.4902	182.1303 (83)

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Total gains 980.9023 1128.1187 1241.7491 1329.5826 1378.2980 1344.7418 1289.7361 1217.2400 1159.1260 1057.6515 960.6118 930.0369 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	45.8477	45.9520	46.0569	46.5883	46.6961	47.2425	47.2425	47.3533	47.0224	46.6961	46.4811	46.2680
alpha	4.0565	4.0635	4.0705	4.1059	4.1131	4.1495	4.1495	4.1569	4.1348	4.1131	4.0987	4.0845
util living area	0.9693	0.9467	0.9058	0.8250	0.6943	0.5232	0.3851	0.4234	0.6313	0.8524	0.9481	0.9744 (86)
Living	19.9387	20.1247	20.3657	20.6273	20.8095	20.8983	20.9195	20.9167	20.8662	20.6316	20.2397	19.8997
Non living	18.9378	19.1719	19.4712	19.7918	19.9975	20.0937	20.1105	20.1106	20.0629	19.8051	19.3259	18.8940
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.4571	20.1247	20.3657	20.6273	20.8095	20.8983	20.9195	20.9167	20.8662	20.6316	20.2397	20.0536 (87)
Th 2	20.1870	20.1887	20.1904	20.1989	20.2006	20.2091	20.2091	20.2108	20.2057	20.2006	20.1972	20.1938 (88)
util rest of house	0.9640	0.9380	0.8906	0.7980	0.6512	0.4650	0.3179	0.3538	0.5725	0.8234	0.9381	0.9700 (89)
MIT 2	19.6890	19.1719	19.4712	19.7918	19.9975	20.0937	20.1105	20.1106	20.0629	19.8051	19.3259	19.1282 (90)
Living area fraction									fLA = Living area / (4) =			0.5125 (91)
MIT	20.0827	19.6603	19.9297	20.2201	20.4137	20.5061	20.5251	20.5238	20.4746	20.2287	19.7943	19.6025 (92)
Temperature adjustment												0.0000
adjusted MIT	20.0827	19.6603	19.9297	20.2201	20.4137	20.5061	20.5251	20.5238	20.4746	20.2287	19.7943	19.6025 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9632	0.9313	0.8851	0.7988	0.6632	0.4872	0.3450	0.3817	0.5931	0.8245	0.9322	0.9661 (94)
Useful gains	944.8270	1050.6668	1099.1085	1062.0248	914.0429	655.2249	444.9597	464.6589	687.4838	872.0563	895.4974	898.5099 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	1858.8016	1734.4427	1574.4985	1312.0223	1007.6086	675.0537	448.6336	470.2366	732.0122	1113.4184	1474.6927	1797.5503 (97)
Space heating kWh	679.9971	459.4974	353.6902	179.9982	69.6129	0.0000	0.0000	0.0000	0.0000	179.5734	417.0206	668.8861 (98a)
Space heating requirement - total per year (kWh/year)												3008.2758
Solar heating kWh	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	679.9971	459.4974	353.6902	179.9982	69.6129	0.0000	0.0000	0.0000	0.0000	179.5734	417.0206	668.8861 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												3008.2758
Space heating per m2										(98c) / (4) =		22.4047 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												365.6366 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												60.0000 (208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating requirement	679.9971	459.4974	353.6902	179.9982	69.6129	0.0000	0.0000	0.0000	0.0000	179.5734	417.0206	668.8861 (98)
Space heating efficiency (main heating system 1)	365.6366	365.6366	365.6366	365.6366	365.6366	0.0000	0.0000	0.0000	0.0000	365.6366	365.6366	365.6366 (210)
Space heating fuel (main heating system)	185.9762	125.6705	96.7327	49.2287	19.0388	0.0000	0.0000	0.0000	0.0000	49.1125	114.0533	182.9374 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating requirement	292.2393	242.2730	214.1748	152.5991	113.2989	98.3116	95.4346	117.3860	153.4890	212.7280	258.8565	288.9571 (64)
Efficiency of water heater (217)m	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142	198.9142 (216)
Fuel for water heating, kWh/month	146.9173	121.7978	107.6720	76.7160	56.9587	49.4241	47.9778	59.0134	77.1634	106.9446	130.1348	145.2672 (219)
Space cooling fuel requirement (221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	38.4857	34.7613	38.4857	37.2442	38.4857	37.2442	38.4857	38.4857	37.2442	38.4857	37.2442	38.4857 (231)
Lighting	32.1764	25.8131	23.2418	17.0280	13.1529	10.7460	11.9985	15.5961	20.2578	26.5793	30.0213	33.0707 (232)
Electricity generated by PVs (Appendix M) (negative quantity)	-63.7894	-91.8511	-132.9236	-147.1521	-153.8524	-140.6268	-138.8391	-132.4315	-119.5097	-103.3813	-70.3499	-54.7903 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)	-212.5732	-192.0016	-212.5732	-205.7160	-212.5732	-205.7160	-212.5732	-212.5732	-205.7160	-212.5732	-205.7160	-212.5732 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235c)
Electricity generated by PVs (Appendix M) (negative quantity)	-26.3663	-58.5928	-123.2996	-196.2375	-270.3766	-276.4578	-272.7336	-227.1617	-162.1239	-88.9412	-36.5878	-20.6547 (233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)	-91.1028	-82.2864	-91.1028	-88.1640	-91.1028	-88.1640	-91.1028	-91.1028	-88.1640	-91.1028	-88.1640	-91.1028 (234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235d)
Annual totals kWh/year												
Space heating fuel - main system 1												822.7502 (211)
Space heating fuel - main system 2												0.0000 (213)
Space heating fuel - secondary												0.0000 (215)
Efficiency of water heater												198.9142
Water heating fuel used												1125.9872 (219)
Space cooling fuel												0.0000 (221)

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Electricity for pumps and fans: (BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6820)	
mechanical ventilation fans (SFP = 0.6820)	373.1382 (230a)
pump for solar water heating	80.0000 (230g)
Total electricity for the above, kWh/year	453.1382 (231)
Electricity for lighting (calculated in Appendix L)	259.6820 (232)
Energy saving/generation technologies (Appendices M ,N and Q)	
PV generation	-3109.0309 (233)
Wind generation	-3575.5408 (234)
Hydro-electric generation (Appendix N)	0.0000 (235a)
Electricity generated - Micro CHP (Appendix N)	0.0000 (235)
Appendix Q - special features	
Energy saved or generated	-0.0000 (236)
Energy used	0.0000 (237)
Total delivered energy for all uses	-4023.0141 (238)

10a. Fuel costs - using Table 12 prices

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year
Space heating - main system 1	822.7502	16.4900	135.6715 (240)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1125.9872	16.4900	185.6753 (247)
Energy for instantaneous electric shower(s)	0.0000	16.4900	0.0000 (247a)
Pumps, fans and electric keep-hot	373.1382	16.4900	61.5305 (249)
Pump for solar water heating	80.0000	16.4900	13.1920 (249)
Energy for lighting	259.6820	16.4900	42.8216 (250)
Additional standing charges			0.0000 (251)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1349.4972	16.4900	-222.5321
PV Unit electricity exported	-1759.5337	5.5900	-98.3579
Total			-320.8900 (252)
Wind Turbine electricity used in dwelling	-2502.8785	16.4900	-412.7247
Wind Turbine electricity exported	-1072.6622	5.5900	-59.9618
Total			-472.6865 (252)
Total energy cost			-354.6857 (255)

11a. SAP rating - Individual heating systems

Energy cost deflator (Table 12):		0.3600 (256)
Energy cost factor (ECF)	[(255) x (256)] / [(4) + 45.0] =	-0.7123 (257)
SAP value		111.5457
SAP rating (Section 12)		112 (258)
SAP band		A

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	822.7502	0.1562	128.4945 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	1125.9872	0.1453	163.5512 (264)
Space and water heating			292.0457 (265)
Pumps, fans and electric keep-hot	453.1382	0.1387	62.8558 (267)
Energy for lighting	259.6820	0.1443	37.4801 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1349.4972	0.1349	-182.0948
PV Unit electricity exported	-1759.5337	0.1243	-218.6627
Total			-400.7575 (269)
Wind Turbine electricity used in dwelling	-2502.8785	0.1387	-347.1801
Wind Turbine electricity exported	-1072.6622	0.1387	-148.7915
Total			-495.9716 (269)
Total CO2, kg/year			-504.3474 (272)
CO2 emissions per m2			-3.7600 (273)
EI value			103.7699
EI rating			104 (274)
EI band			A

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF EPC COSTS, EMISSIONS AND PRIMARY ENERGY FOR IMPROVED DWELLING

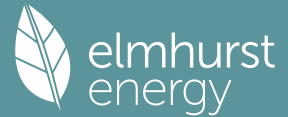
1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)
Main dwelling	134.2700 (1b)	x 3.3400 (2b)	= 448.4618 (1b) - (3b)
Ground floor			(4)
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	134.2700		448.4618 (5)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	

2. Ventilation rate

m3 per hour

Full SAP Calculation Printout



Number of open chimneys
Number of open flues
Number of chimneys / flues attached to closed fire
Number of flues attached to solid fuel boiler
Number of flues attached to other heater
Number of blocked chimneys
Number of intermittent extract fans
Number of passive vents
Number of flueless gas fires

0 * 80 = 0.0000 (6a)
0 * 20 = 0.0000 (6b)
1 * 10 = 10.0000 (6c)
0 * 20 = 0.0000 (6d)
0 * 35 = 0.0000 (6e)
0 * 20 = 0.0000 (6f)
0 * 10 = 0.0000 (7a)
0 * 10 = 0.0000 (7b)
0 * 40 = 0.0000 (7c)

Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) = 10.0000 / (5) = 0.0223 (8)
Pressure test
Pressure Test Method
Measured/design AP50
Infiltration rate
Number of sides sheltered
Blower Door
Shelter factor (20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor (21) = (18) x (20) = 0.0723 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.7000	5.4000	5.4000	4.8000	4.8000	4.2000	4.2000	4.1000	4.4000	5.0000	5.1000	5.6000 (22)
Wind factor	1.4250	1.3500	1.3500	1.2000	1.2000	1.0500	1.0500	1.0250	1.1000	1.2500	1.2750	1.4000 (22a)
Adj infilt rate	0.1030	0.0976	0.0976	0.0868	0.0868	0.0759	0.0759	0.0741	0.0795	0.0904	0.0922	0.1012 (22b)
Balanced mechanical ventilation with heat recovery												
If mechanical ventilation												0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												84.6000 (23c)
Effective ac	0.1800	0.1746	0.1746	0.1638	0.1638	0.1529	0.1529	0.1511	0.1565	0.1674	0.1692	0.1782 (25)

3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K
Main dwelling							
Door			4.4500	0.9615	4.2788		(27)
Window			20.9300	0.9615	20.1250		(27)
Door			1.7100	1.0000	1.7100		(26)
Ground Floor			134.2700	0.1100	14.7697	110.0000	14769.7000 (28a)
External Wall 1 - Stone	136.6100	27.0900	109.5200	0.1400	15.3328	9.0000	985.6800 (29a)
External Roof 1 - Pitched	155.0400		155.0400	0.1400	21.7056	9.0000	1395.3600 (30)
Total net area of external elements Aum(A, m2)			425.9200				(31)
Fabric heat loss, W/K = Sum (A x U)				(26)...(30) + (32) =	77.9219		(33)
Main dwelling							
Party Wall 1			5.2800	0.0000	0.0000	20.0000	105.6000 (32)
Internal Wall 1 - GF			242.5100			9.0000	2182.5900 (32c)

Heat capacity Cm = Sum(A x k)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K
(28)...(30) + (32) + (32a)...(32e) = 19438.9300 (34)
List of Thermal Bridges
144.7749 (35)

K1 Element	Length	Psi-value	Total
E2 Other lintels (including other steel lintels)	19.1200	0.0840	1.6061
E3 Sill	10.8000	0.0340	0.3672
E4 Jamb	36.2000	0.0430	1.5566
E5 Ground floor (normal)	54.4600	0.0210	1.1437
E11 Eaves (insulation at rafter level)	43.4400	0.0400	1.7376
E13 Gable (insulation at rafter level)	12.7200	0.0240	0.3053
E16 Corner (normal)	7.6500	0.0300	0.2295
R4 Ridge (vaulted ceiling)	24.3600	0.1200	2.9232
E18 Party wall between dwellings	5.1000	0.0400	0.2040
P1 Party wall - Ground floor	5.2800	0.3200	1.6896
P5 Party wall - Roof (insulation at rafter level)	6.3600	0.4800	3.0528
Thermal bridges (Sum(L x Psi) calculated using Appendix K)			14.8155 (36)
Point Thermal bridges			0.0000 (36a)
Total fabric heat loss			(33) + (36) + (36a) = 92.7375 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)
(38)m
Heat transfer coeff
Average = Sum(39)m / 12 =

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(38)m	26.6424	25.8399	25.8399	24.2350	24.2350	22.6300	22.6300	22.3625	23.1650	24.7699	25.0374	26.3749 (38)
Heat transfer coeff	119.3798	118.5774	118.5774	116.9724	116.9724	115.3675	115.3675	115.1000	115.9025	117.5074	117.7749	119.1123 (39)
Average = Sum(39)m / 12 =												117.2176

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
HLP	0.8891	0.8831	0.8831	0.8712	0.8712	0.8592	0.8592	0.8572	0.8632	0.8752	0.8771	0.8871 (40)
HLP (average)												0.8730
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

4. Water heating energy requirements (kWh/year)

Assumed occupancy
Hot water usage for mixer showers
Hot water usage for baths
Hot water usage for other uses
Average daily hot water use (litres/day)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Hot water usage for mixer showers	72.9317	71.8357	70.2386	67.1828	64.9277	62.4128	60.9834	62.5684	64.3059	67.0061	70.1275	72.6523 (42a)
Hot water usage for baths	31.4863	31.0186	30.3601	29.1460	28.2368	27.2287	26.6842	27.3381	28.0501	29.1288	30.3679	31.3798 (42b)
Hot water usage for other uses	44.3829	42.7690	41.1551	39.5412	37.9272	36.3133	36.3133	37.9272	39.5412	41.1551	42.7690	44.3829 (42c)
Average daily hot water use (litres/day)												136.7815 (43)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Daily hot water use	148.8009	145.6234	141.7538	135.8699	131.0918	125.9549	123.9809	127.8338	131.8971	137.2900	143.2645	148.4151 (44)
Energy conte	235.6643	207.3657	217.8704	185.9992	176.4747	154.8762	149.9442	158.2849	162.6424	186.3012	204.1065	232.3821 (45)
Energy content (annual)										Total = Sum(45)m =		2271.9117
Distribution loss (46)m = 0.15 x (45)m												
35.3496	31.1049	32.6806	27.8999	26.4712	23.2314	22.4916	23.7427	24.3964	27.9452	30.6160	34.8573 (46)	
Water storage loss (or HIU loss):												
Store volume												150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):												1.9900 (48)
Temperature factor from Table 2b												0.5400 (49)

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Enter (49) or (54) in (55)												1.0746 (55)
Total storage loss	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (56)
If cylinder contains dedicated solar storage	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (57)
Primary loss	23.2624	21.0112	21.8667	15.7584	10.4681	9.9053	10.2355	11.1660	17.1091	21.8667	22.5120	23.2624 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month	292.2393	258.4657	273.0496	233.9956	220.2554	197.0195	193.4922	202.7634	211.9895	241.4804	258.8565	288.9571 (62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)
Aperture area of solar collector												3.0000 (H1)
Zero-loss collector efficiency												0.8000 (H2)
Collector linear heat loss coefficient												1.8000 (H3)
Collector 2nd order heat loss coefficient												0.0000 (H4)
Collector loop efficiency												0.9000 (H5)
Incidence angle modifier												1.0000 (H6)
Overshading factor												0.8000 (H8)
Overall heat loss coefficient of system												6.5000 (H10)
Heat loss coefficient of collector loop												3.9667 (H11)
Dedicated solar storage volume												75.0000 (H12)
Effective solar volume												75.0000 (H14)
Reference volume												225.0000 (H15)
Storage tank correction coefficient												1.3161 (H16)
Heat delivered to hot water												770.4241 (H24)
Heat delivered to space heating												0.0000 (H29)
Solar input												770.4241
Solar input	-7.3048	-25.7498	-71.3961	-96.6532	-115.9886	-117.2078	-105.4749	-101.2346	-75.1711	-43.2748	-10.9684	-0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h	284.9345	232.7160	201.6535	137.3424	104.2668	79.8117	88.0173	101.5288	136.8184	198.2056	247.8881	288.9571 (64)
												Total per year (kWh/year) = Sum(64)m = 2102.1402 (64)
Electric shower(s)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)
												Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m = 0.0000 (64a)
Heat gains from water heating, kWh/month	123.6184	109.8291	116.5853	100.2418	93.7024	85.2110	84.6949	88.2126	93.5563	106.0886	111.6654	122.5271 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(66)m	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	174.3165	(66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	36.7607	32.6505	26.5532	20.1025	15.0268	12.6863	13.7080	17.8181	23.9155	30.3662	35.4418	37.7824	(67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	451.9355	456.6252	444.8073	419.6486	387.8901	358.0417	338.1011	333.4114	345.2293	370.3880	402.1465	431.9949	(68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	55.3369	(69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(70)
Losses e.g. evaporation (negative values) (Table 5)	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	-116.2110	(71)
Water heating gains (Table 5)	166.1537	163.4362	156.7007	139.2248	125.9441	118.3486	113.8372	118.5653	129.9393	142.5921	155.0909	164.6869	(72)
Total internal gains	768.2923	766.1543	741.5036	692.4182	642.3035	602.5190	579.0887	583.2373	612.5264	656.7888	706.1216	747.9066	(73)

6. Solar gains

[Jan]	Area m2	Solar flux Table 6a W/m2	g Specific data or Table 6b	FF Specific data or Table 6c	Access factor Table 6d	Gains W
South	4.4500	58.8467	0.6300	0.7000	0.7700	80.0301 (78)
North	12.5700	14.1647	0.6300	0.7000	0.7700	54.4144 (74)
South	7.0000	58.8467	0.6300	0.7000	0.7700	125.8901 (78)
West	1.3600	26.5321	0.6300	0.7000	0.7700	11.0276 (80)

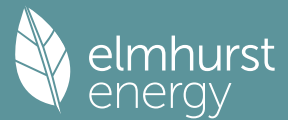
Solar gains	271.3623	397.0642	542.6195	712.6389	785.0878	867.4216	758.4583	721.7457	629.3924	457.1585	313.0134	223.7643 (83)
Total gains	1039.6546	1163.2185	1284.1231	1405.0571	1427.3912	1469.9405	1337.5470	1304.9830	1241.9188	1113.9473	1019.1350	971.6709 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	45.2313	45.5374	45.5374	46.1622	46.1622	46.8044	46.8044	46.9131	46.5883	45.9520	45.8477	45.3329
alpha	4.0154	4.0358	4.0358	4.0775	4.0775	4.1203	4.1203	4.1275	4.1059	4.0635	4.0565	4.0222
util living area	0.9452	0.9193	0.8720	0.7888	0.6769	0.5008	0.4155	0.4081	0.5734	0.7858	0.9043	0.9512 (86)
Living	20.1886	20.3173	20.4923	20.6828	20.8175	20.9004	20.9158	20.9169	20.8846	20.7358	20.4699	20.1816
Non living	19.2445	19.4063	19.6177	19.8487	19.9973	20.0879	20.1008	20.1034	20.0723	19.9122	19.6015	19.2384
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.5849	20.3173	20.4923	20.6828	20.8175	20.9004	20.9158	20.9169	20.8846	20.7358	20.4699	20.2961 (87)
Th 2	20.1768	20.1819	20.1819	20.1921	20.1921	20.2023	20.2023	20.2040	20.1989	20.1887	20.1870	20.1785 (88)
util rest of house	0.9352	0.9055	0.8512	0.7579	0.6329	0.4455	0.3507	0.3416	0.5124	0.7453	0.8850	0.9418 (89)
MIT 2	19.8051	19.4063	19.6177	19.8487	19.9973	20.0879	20.1008	20.1034	20.0723	19.9122	19.6015	19.4078 (90)
Living area fraction												0.5125 (91)
MIT	20.2048	19.8732	20.0660	20.2762	20.4177	20.5044	20.5185	20.5204	20.4886	20.3343	20.0466	19.8631 (92)
Temperature adjustment												0.0000
adjusted MIT	20.2048	19.8732	20.0660	20.2762	20.4177	20.5044	20.5185	20.5204	20.4886	20.3343	20.0466	19.8631 (93)

8. Space heating requirement

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	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Utilisation	0.9356	0.8995	0.8480	0.7613	0.6456	0.4668	0.3766	0.3683	0.5351	0.7531	0.8812	0.9376	(94)
Useful gains	972.6518	1046.3688	1088.8891	1069.6537	921.5405	686.1359	503.7828	480.5720	664.4895	838.8692	898.0922	911.0603	(95)
Ext temp.	6.4000	6.7000	7.8000	9.4000	11.8000	14.4000	16.1000	16.3000	14.5000	11.9000	9.2000	6.8000	(96)
Heat loss rate W	1648.0133	1562.0480	1454.4682	1272.2192	1008.0322	704.2445	509.7517	485.7662	694.0949	991.0951	1277.4583	1555.9740	(97)
Space heating kWh	502.4690	346.5364	271.9908	145.8472	64.3499	0.0000	0.0000	0.0000	0.0000	113.2560	273.1436	479.8158	(98a)
Space heating requirement - total per year (kWh/year)												2197.4086	
Solar heating kWh	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	(98b)
Solar heating contribution - total per year (kWh/year)												0.0000	
Space heating kWh	502.4690	346.5364	271.9908	145.8472	64.3499	0.0000	0.0000	0.0000	0.0000	113.2560	273.1436	479.8158	(98c)
Space heating requirement after solar contribution - total per year (kWh/year)												2197.4086	
Space heating per m2										(98c) / (4) =		16.3656	(99)

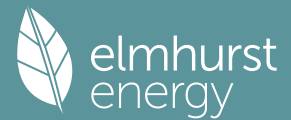
9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)													0.0000	(201)
Fraction of space heat from main system(s)													1.0000	(202)
Efficiency of main space heating system 1 (in %)													364.9749	(206)
Efficiency of main space heating system 2 (in %)													0.0000	(207)
Efficiency of secondary/supplementary heating system, %													60.0000	(208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
Space heating requirement	502.4690	346.5364	271.9908	145.8472	64.3499	0.0000	0.0000	0.0000	0.0000	113.2560	273.1436	479.8158	(98)	
Space heating efficiency (main heating system 1)	364.9749	364.9749	364.9749	364.9749	364.9749	0.0000	0.0000	0.0000	0.0000	364.9749	364.9749	364.9749	(210)	
Space heating fuel (main heating system)	137.6722	94.9480	74.5232	39.9609	17.6313	0.0000	0.0000	0.0000	0.0000	31.0312	74.8390	131.4654	(211)	
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(212)	
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(213)	
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(215)	
Water heating														
Water heating requirement	284.9345	232.7160	201.6535	137.3424	104.2668	79.8117	88.0173	101.5288	136.8184	198.2056	247.8881	288.9571	(64)	
Efficiency of water heater (217)m	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	198.7969	(216)	
Fuel for water heating, kWh/month	143.3295	117.0622	101.4370	69.0868	52.4489	40.1474	44.2750	51.0716	68.8232	99.7026	124.6942	145.3530	(219)	
Space cooling fuel requirement (221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(221)	
Pumps and Fa	38.4857	34.7613	38.4857	37.2442	38.4857	37.2442	38.4857	38.4857	37.2442	38.4857	37.2442	38.4857	(231)	
Lighting	32.1764	25.8131	23.2418	17.0280	13.1529	10.7460	11.9985	15.5961	20.2578	26.5793	30.0213	33.0707	(232)	
Electricity generated by PVs (Appendix M) (negative quantity) (233a)m	-78.6727	-99.1903	-140.2985	-155.3716	-157.8940	-148.0493	-142.2367	-139.8630	-129.3219	-113.2657	-83.1393	-65.8264	(233a)	
Electricity generated by wind turbines (Appendix M) (negative quantity) (234a)m	-212.5732	-192.0016	-212.5732	-205.7160	-212.5732	-205.7160	-212.5732	-212.5732	-205.7160	-212.5732	-205.7160	-212.5732	(234a)	
Electricity generated by hydro-electric generators (Appendix M) (negative quantity) (235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235a)	
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation) (235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235c)	
Electricity generated by PVs (Appendix M) (negative quantity) (233b)m	-40.2593	-71.3261	-145.2364	-234.6530	-297.3464	-340.5635	-298.7341	-274.0724	-202.4386	-113.0448	-52.8303	-29.9404	(233b)	
Electricity generated by wind turbines (Appendix M) (negative quantity) (234b)m	-91.1028	-82.2864	-91.1028	-88.1640	-91.1028	-88.1640	-91.1028	-91.1028	-88.1640	-91.1028	-88.1640	-91.1028	(234b)	
Electricity generated by hydro-electric generators (Appendix M) (negative quantity) (235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235b)	
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation) (235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235d)	
Annual totals kWh/year														
Space heating fuel - main system 1													602.0711	(211)
Space heating fuel - main system 2													0.0000	(213)
Space heating fuel - secondary													0.0000	(215)
Efficiency of water heater													198.7969	
Water heating fuel used													1057.4313	(219)
Space cooling fuel													0.0000	(221)
Electricity for pumps and fans:														
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6820)														
mechanical ventilation fans (SFP = 0.6820)													373.1382	(230a)
pump for solar water heating													80.0000	(230g)
Total electricity for the above, kWh/year													453.1382	(231)
Electricity for lighting (calculated in Appendix L)													259.6820	(232)
Energy saving/generation technologies (Appendices M ,N and Q)														
PV generation													-3553.5745	(233)
Wind generation													-3575.5408	(234)
Hydro-electric generation (Appendix N)													0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)													0.0000	(235)
Appendix Q - special features														
Energy saved or generated													-0.0000	(236)
Energy used													0.0000	(237)
Total delivered energy for all uses													-4756.7927	(238)

10a. Fuel costs - using BEDF prices (595)

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year	
Space heating - main system 1	602.0711	27.6700	166.5931	(240)
Total CO2 associated with community systems			0.0000	(473)
Water heating (other fuel)			292.5912	(247)
Energy for instantaneous electric shower(s)	1057.4313	27.6700	0.0000	(247a)
Pumps, fans and electric keep-hot	373.1382	27.6700	103.2473	(249)
Pump for solar water heating	80.0000	27.6700	22.1360	(249)
Energy for lighting	259.6820	27.6700	71.8540	(250)
Additional standing charges			0.0000	(251)
Energy saving/generation technologies				

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PV Unit electricity used in dwelling	-1453.1294	27.6700	-402.0809
PV Unit electricity exported	-2100.4452	5.8100	-122.0359
Total			-524.1168 (252)
Wind Turbine electricity used in dwelling	-2502.8785	27.6700	-692.5465
Wind Turbine electricity exported	-1072.6622	5.8100	-62.3217
Total			-754.8682 (252)
Total energy cost			-622.5633 (255)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	602.0711	0.1562	94.0258 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	1057.4313	0.1460	154.3447 (264)
Space and water heating			248.3706 (265)
Pumps, fans and electric keep-hot	453.1382	0.1387	62.8558 (267)
Energy for lighting	259.6820	0.1443	37.4801 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1453.1294	0.1355	-196.9039
PV Unit electricity exported	-2100.4452	0.1252	-262.8857
Total			-459.7896 (269)
Wind Turbine electricity used in dwelling	-2502.8785	0.1387	-347.1801
Wind Turbine electricity exported	-1072.6622	0.1387	-148.7915
Total			-495.9716 (269)
Total CO2, kg/year			-607.0547 (272)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year
Space heating - main system 1	602.0711	1.5781	950.1538 (275)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1057.4313	1.5399	1628.3829 (278)
Space and water heating			2578.5366 (279)
Pumps, fans and electric keep-hot	453.1382	1.5128	685.5074 (281)
Energy for lighting	259.6820	1.5338	398.3089 (282)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1453.1294	1.5008	-2180.9135
PV Unit electricity exported	-2100.4452	0.4594	-964.8968
Total			-3145.8104 (283)
Wind Turbine electricity used in dwelling	-2502.8785	1.5128	-3786.3546
Wind Turbine electricity exported	-1072.6622	0.5128	-550.0612
Total			-4336.4158 (283)
Total Primary energy kWh/year			-3819.8732 (286)